FOLLOW MY VOICE: SELF-SIMILAR AND NEUTRAL VOICES FOR GAZE-CONTINGENT OBJECT SEARCH IN VIRTUAL REALITY

by

PEDRO POVEDA QUEVEDO
B.Sc. in Computer Science

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Major Professor: Dr. Roshan Venkatakrishnan

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GENERATIVE ARTIFICIAL INTELLIGENCE USE DISCLOSURE

This first-pass manuscript was prepared with assistance from OpenAI Codex for source organization, drafting, implementation-to-protocol reconciliation, and LaTeX production. Claude performed separate language and methods reviews; their findings and the changes made in response are recorded with the source project. The author must review the manuscript, verify its citations and claims, and approve the final disclosure before submission. This working disclosure does not assert that author validation or committee approval has already occurred.

ABSTRACT

Directing users' attention to objects of interest has long been a concern in virtual reality, particularly when relevant objects lie outside the user's current field of view [15, 24]. Research in augmented reality has shown that auditory guidance, including verbal hints, can support object search [44]. Spoken assistance can connect spatial instructions with objects in the surrounding environment, guiding users toward a region and then the intended object. The experience of receiving assistance can also be influenced by the tone and voice identity of spoken instructions. Research on virtual collaboration has found that self-similar voices can increase likability and believability while also eliciting greater eeriness [17]. As interactive agents take on a more active role in guiding users through virtual environments, the experience of receiving their assistance becomes an important design consideration. These findings motivate examining how self-similarity affects users when the voice serves as a spatial guide. This thesis compares gaze-contingent guidance spoken in a synthetic voice resembling the participant's own (self-similar) with guidance spoken in a neutral voice selected by the participant from a supplied catalog.

Follow My Voice will investigate this comparison in a within-participant virtual-reality user study. Participants will complete one practice trial in their first assigned voice and two counterbalanced blocks of 40 measured trials. Seated participants will search colored geometric objects on eight surrounding vertical planes, including locations initially outside their field of view. Initially, participants will not be informed of which object they should find. Instead, an agent that uses the participant's gaze data as context will guide them toward it. Participants will select the target by pointing the right controller at the object and pressing the trigger. Both trial conditions will use the same hint guidance to help participants find the object.

The study will compare active search-completion time, perceived workload, motivation, and agent experience. Condition-specific questionnaires will be administered after both blocks.

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PREFACE

This is an advisor-review draft revised October 23, 2026. It expands the project's current Overleaf manuscript and bibliography into a thesis using the supplied UCF ETD LuaLaTeX template. It is not a completed empirical report. Present-tense system descriptions refer to the documented implementation snapshot; future-tense methods describe the proposed study. Engineering checks, planned measurements, and participant outcomes are kept separate.

The source snapshot is Overleaf revision `f9f49e4`. The runtime alignment audit dated October 20 is the accompanying implementation record. Appendix E separately documents an October 23 inspection of three working implementation files, with their hashes and uncommitted status retained. That inspection does not update the deployment claims of the October 20 audit. Later software changes require a new reconciliation before submission. Exact pre-cue geometry, balanced vertical allocation, final sample size, and pilot acceptance rules remain explicit readiness requirements. Qualtrics synchronization remains on hold. The author has supplied the previous degree, B.Sc. in Computer Science, and identified Dr. Roshan Venkatakrishnan as major professor. The degree-granting institution and year, any remaining administrative details, and the final AI disclosure still require author confirmation before submission.

The structural reference is Hoang Bao Truong's 2026 UCF thesis, *Using Inferences from Natural Language Narratives to Improve Video Question Answering*. Its chapter hierarchy, front matter, detailed methods, diagnostic discussion, and appendices inform presentation only. Its prose, experiments, and findings are not evidence for this study.

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LIST OF ABBREVIATIONS

AI Artificial intelligence

API Application programming interface

AR Augmented reality

ARM64 64-bit Arm architecture

CSV Comma-separated values

EQ1 Engineering question 1

ETD Electronic thesis or dissertation

ID Identifier

IMI Intrinsic Motivation Inventory

JSON JavaScript Object Notation

NASA-TLX National Aeronautics and Space Administration Task Load Index

PDF Portable Document Format

RQ1–RQ3 Research questions 1–3

SSQ Simulator Sickness Questionnaire

SVG Scalable Vector Graphics

TLX Task Load Index (see NASA-TLX)

UCF University of Central Florida

VR Virtual reality

XR Extended reality

CHAPTER 1: INTRODUCTION

1.1 Motivation and Background

Directing users' attention to objects of interest has long been a concern in virtual reality (VR), particularly when relevant content lies outside the user's current field of view. An immersive environment can surround a user with information, yet only part of that environment is visible at any moment. Grogorick et al. address this problem through subtle gaze guidance across an immersive scene, while HiveFive uses swarm motion to direct attention while preserving immersion [15, 24]. These approaches illustrate a recurring design problem: how can a system help users locate relevant content while allowing them to remain engaged with the surrounding environment?

1.1.1 Guiding Users Toward Objects of Interest

Object search gives attention guidance a practical purpose. Finding the relevant object is often a prerequisite for following an instruction or interacting with an environment. Binetti et al. motivate augmented-reality (AR) search assistance through maintenance, repair, instruction, and training, where a person must locate a target in the surrounding scene [5]. Zhang et al. investigate the concrete example of finding a specified book among other books [44]. Such tasks let researchers study how users connect a cue with its intended object without reproducing an entire workplace activity.

Chiossi et al.'s *Searching Across Realities* uses object search to examine how the presentation of physical and virtual content affects search efficiency and attention allocation [9]. This provides a methodological rationale for treating object search as a controlled task with behavioral and gaze measures. Their comparison concerns augmented reality and augmented virtuality, rather than

voice identity; it motivates specifying the visual environment carefully while the present study varies the guiding voice.

When an object begins outside the field of view, assistance must first help the user orient toward a relevant region. Once that region becomes visible, the user still needs to distinguish the intended object from nearby distractors. This distinction between orientation and local search motivates considering guidance as an unfolding interaction rather than a single cue. Visual techniques offer ways to direct attention [15, 24]; auditory guidance provides another way to communicate where to look as that interaction develops.

1.1.2 From Auditory Cues to Spoken Assistance

Research on auditory guidance supports using sound to help users locate objects, although its benefits depend on the task and cue design. In head-mounted AR, Binetti et al. combined visual arrows with spatialized alarm sounds to guide users toward out-of-view targets [5]. Adding sound mitigated response-time costs when visual cues and targets occupied different depth planes, but did not remove the corresponding accuracy cost. The study supports considering an auditory channel alongside visual guidance; it does not establish that speech alone is superior to visual cues.

Spoken hints provide a more direct precedent for the present task. Zhang et al.'s AR book-search study used an audio instruction identifying an approximate shelf [44]. Hint conditions improved performance relative to the control in the sub-study without post-task feedback, while advantages of combined visual and audio hints depended on task and feedback exposure. Their verbal cue was delivered at task onset rather than continuously in response to gaze. The study therefore motivates spoken search assistance without establishing the effectiveness of the more adaptive interaction proposed here.

Gaze-contingent sonification and staged auditory guidance offer complementary precedents for adapting assistance to search progress. Losing et al. varied a tone with gaze-to-target distance, and Kwok et al. combined initial spatial-audio orientation with finer guidance [27, 23]. These studies used desktop displays rather than immersive headsets, and their cues should not be equated with conversational speech. Their relevance is the connection between what the user is currently looking at and what information the system supplies next.

Follow My Voice applies this idea through short spoken directions: first orient toward a target region, then refine the search within it. The agent knows the target, while the participant initially does not know which object to select. Speech consequently has an ongoing instructional role, progressively connecting a direction with an intended object. This provides a setting in which the user repeatedly encounters the guiding voice, raising a further question: does hearing that guidance in a voice resembling one's own change the experience or effectiveness of receiving help?

1.1.3 Self-Similar Voice and the Doppelgänger Connection

A spatial instruction conveys both a direction and a voice identity. The words "look to your right" specify an action, while the voice delivering them may be familiar, appealing, or uncomfortable. Self-similar synthesis uses recorded speech to produce a voice resembling the participant. Its relevance to guidance therefore extends beyond whether the instruction is intelligible: the resemblance may change how the participant experiences the agent giving that instruction.

Guo et al.'s *Collaborating with my Doppelgänger* provides a direct motivation for investigating this possibility [17]. Their 25-participant VR study crossed self-similar appearance and voice during a jigsaw puzzle co-solving task with a virtual character. Self-similar voices increased likability and believability ratings while also increasing eeriness; believability additionally depended on the interaction between appearance and voice. These findings show why favorable impressions and

discomfort should be examined separately. They do not establish that self-similar speech improves object-search speed or that the same responses will occur when the guide has no visible body.

The present study carries that question from an embodied collaborator to a voice serving as a spatial guide. In the puzzle task, participants interacted with a visible character; here, they will receive directions toward an initially unnamed object from an agent without a visible body. This change in role matters because the participant must use the speech to determine where to look and what to select. Whether self-resemblance makes those instructions easier to follow, more welcome, or more uncomfortable remains an empirical question. Research on self-similar avatar voices in an educational game supplies additional motivation through selected engagement and identification outcomes, while also underscoring that findings depend on the task and measures [20].

For designers, the comparison of interest is whether self-similar synthesis offers value beyond a voice the user already likes. Follow My Voice therefore compares a self-similar synthetic voice with a neutral voice selected by the participant from a supplied catalog. Throughout this thesis, *neutral voice* names the comparison condition: an existing voice that is not cloned from the participant. The label does not imply an emotionally flat delivery, a lack of preference, or verified absence of perceived resemblance. Voice selection accommodates preference; cloning introduces resemblance to the participant. Both are practical ways to personalize an assistant, so the study asks what self-resemblance contributes when the alternative already reflects the user's choice. Because the voices can differ in acoustic properties as well as resemblance, the comparison concerns these two voice configurations rather than an isolated causal effect of similarity.

Usefulness will be evaluated through both behavior and experience. Recognizing the clone is a manipulation check, not a performance benefit; liking it without searching faster is still an experience result. Likewise, completing efficiently while finding the voice eerie would distinguish task effectiveness from acceptability. The thesis therefore examines completion time, workload,

motivation, and agent experience separately. This approach follows the Doppelgänger findings in allowing favorable and adverse reactions to coexist, without assuming that self-similar guidance will be preferable.

1.1.4 A Controlled Setting for Self-Similar Assistance

Follow My Voice studies this comparison in a surrounding object-search task. A seated participant wears a virtual-reality headset and views colored geometric objects distributed across eight vertical planes. Some targets will initially lie outside the participant's field of view. The application knows which object is the target, while the participant is not given its color or shape at the start. Spoken directions first guide orientation toward the relevant wall and then refine the search within it. The participant aims a right-controller ray and presses the trigger to select the inferred target; gaze itself does not select an object.

The controlled shapes and surrounding layout make starting geometry and trial events easier to specify across the two voice conditions. Object search gives the speech a concrete purpose and observable outcomes, while the initially unnamed target makes guidance consequential throughout the task. Maintenance, training, and everyday retrieval motivate the broader value of search assistance, but transfer to those settings would require further evaluation.

The central comparison varies the assigned voice while the assistance mechanism stays common to both conditions. Gaze contingency and constrained agent decisions deliver relevant instructions but are not additional participant treatments. They provide the shared apparatus for asking whether hearing guidance in one's own synthetic voice changes the efficiency or experience of being guided.

1.2 Problem Statement

Together, the guidance literature and the self-similar voice literature motivate a question that neither establishes on its own: how will self-similar spoken assistance compare with a neutral voice when guiding search for an initially unnamed object? Auditory search studies motivate the use of sound and verbal hints [5, 44], while the Doppelgänger study motivates examining both favorable impressions and eeriness [17]. The unresolved issue is how those voice experiences relate to task performance and receiving repeated spatial instructions in the present setting.

The study therefore compares two practical voice configurations within a common task. It asks whether self-similar guidance changes active search time, workload, motivation, and the experience of the agent relative to neutral-voice guidance, keeping these outcomes separate rather than defining usefulness through a single favorable score. Because the voices can also differ in naturalness and other acoustic properties, it does not isolate resemblance as their only difference.

A credible answer requires appropriate guidance in both conditions and a record of what each participant actually heard. Gaze contingency, the shared decision policy, and playback logging serve that requirement. Independent gaze-quality checks and distinct records of looking, pointing, and selecting protect the interpretation of the interaction [13, 33]. Chapters 3 and 5 describe these controls; the central empirical question remains the value of self-similar voice as assistance.

1.3 Research Objectives and Questions

The primary objective is to evaluate self-similar voice as a form of task assistance, using a neutral voice as the comparison within a shared gaze-contingent system. The primary research question is: will active search-completion time differ between self-similar and neutral voices after accounting

for absolute starting angle and repeated observations within participants? The contrast is two-sided because the current design does not justify assuming a performance advantage for self-similarity.

Secondary questions concern perceived workload and activity experience. Will the two voices differ on the six workload dimensions and their complete unweighted mean? Will they differ on interest/enjoyment, effort/importance, pressure/tension, or value/usefulness? Because these outcomes concern different constructs, the analysis will preserve those distinctions rather than pool them into a broad satisfaction score.

Exploratory questions concern agent perception, reliance, and the relationship between task difficulty and voice experience. Participants may describe a voice as more responsive or humanlike, may rely on guidance more often, or may experience a cloned voice as familiar or eerie. Such reports will be treated as judgments about an interaction, not as evidence that the system possesses intelligence, emotions, awareness, or an inner voice.

A supporting validation requirement is to make the assistance process inspectable so that technical faults do not masquerade as effects of self-similar voice. Can recorded state and speech history explain why a cue was eligible, why it played or was suppressed, and how long it waited? Can evaluation distinguish a genuinely unhelpful instruction from an acceptable alternative wording? These questions concern system validity and readiness; they support the empirical study but do not replace it.

1.3.1 Research-Question Traceability

The questions are assigned stable identifiers so that the completed results and conclusion can answer the same questions introduced here. RQ1 is the primary empirical contrast; RQ2 concerns secondary experience measures; RQ3 is exploratory. EQ1 identifies an engineering question rather

than a fourth participant research question. Table 1.1 connects each question to its evidence and analysis. The identifiers organize the existing plan and do not add a new experimental factor or a directional hypothesis.

Table 1.1: Research-question traceability for the proposed study. No empirical answer is available in this first pass.

Question	Intended contrast	Evidence and reporting destination
RQ1	Does self-similar voice change active completion time relative to a neutral voice?	Trial timing and starting geometry; analysis in Chapter 6, results in Chapter 7.
RQ2	Do self-similar and neutral voices differ in workload or motivation?	Paired block-referenced workload and separate IMI subscales; final retrospective administration.
RQ3	How do voices differ in agent perception, voice experience, reliance, and search behavior?	Custom ratings, agent constructs, debrief, and valid search measures; exploratory voice-by-angle comparison.
EQ1	Can the recorded sequence explain eligibility, delivery, and unhelpful guidance?	State, request, and actual playback records; supporting evaluation in Appendix F.

The voice-similarity manipulation check will be reported before the outcome contrasts to aid interpretation across questions. It is not an individual exclusion gate. RQ3 includes appealing, eerie, and familiar voice experiences and the exploratory variation of the voice contrast with starting angle.

1.4 Scope of the Comparison

The independent variable is voice condition: neutral voice versus self-similar synthetic voice. Both conditions use gaze contingency, the same semantic phrase library, the same local geometry rules, and the same controller-selection mechanics. Consequently, the study will not isolate the benefit of gaze contingency, short-term memory, the cloud decision agent, or synthetic speech in general. Those are shared system properties.

The task is deliberately narrower than unrestricted navigation. Participants remain seated and rotate their head and chair to search surrounding planes. Guidance such as "turn left" concerns orientation within this setup; it does not ask them to walk toward a target. Objects will appear in a virtual room, and the target will initially be unnamed. Conclusions will therefore be bounded to guided search under these conditions.

The use of one practice trial in the first assigned voice is a current protocol decision. It introduces an exposure asymmetry: whichever voice appears first also appears during practice. Counterbalancing distributes that asymmetry across voices, but does not remove learning or carryover. The analysis must retain period and assigned order, and the report must describe the practice assignment explicitly.

Figure 1.1 separates the voice manipulation from the shared task. A difference between conditions would require evidence from the planned study; the diagram specifies the comparison rather than predicting its direction.

1.5 Contributions of This First Pass

The thesis develops an empirical comparison of self-similar voice with a preferred alternative in virtual-reality task assistance. Its conceptual contribution is to distinguish resemblance from preference and to examine performance alongside the favorable or uncomfortable experience of hearing one's own synthetic voice. The study will not assume that greater perceived similarity implies faster search, lower workload, or a more welcome agent.

The methodological contribution is a within-participant design that makes this comparison operational: two counterbalanced voice blocks, a shared hidden-target task, explicit trigger selection, identical nominal starting-angle quotas across blocks, retrospective voice-referenced ques-

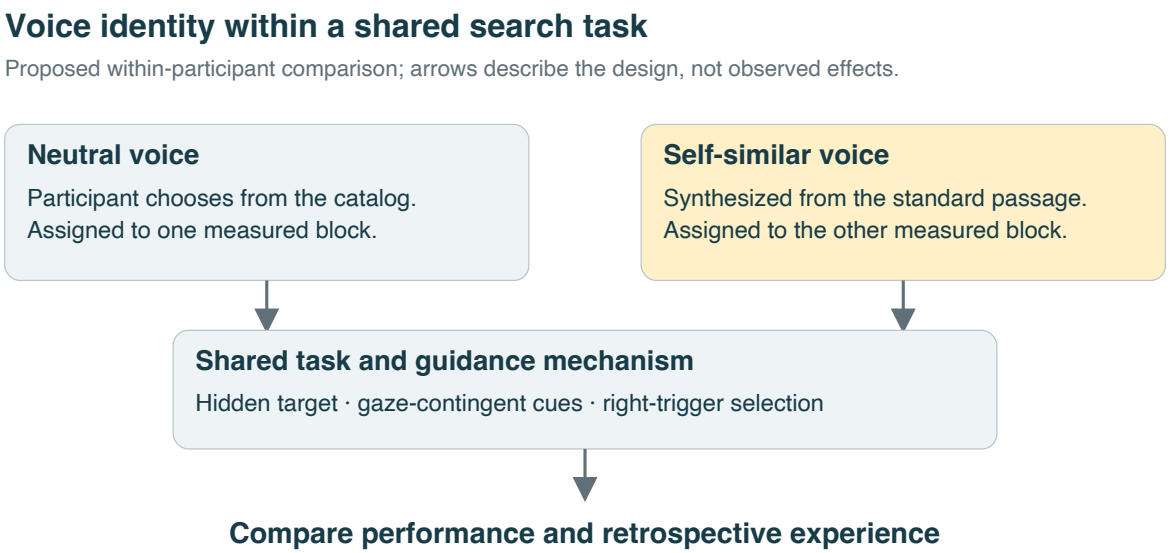


Figure 1.1: Proposed comparison of neutral and self-similar voices within a shared task and guidance mechanism. The schematic identifies the intended manipulation and outcome families; it does not depict observed effects.

tionnaires, and a planned repeated-measures analysis. Manipulation checks and acoustic-quality reporting will help establish what the voice conditions actually delivered before their outcomes are interpreted.

The implemented guidance and recording platform supports that empirical contribution. Local spatial eligibility, constrained agent decisions, prepared speech, and synchronized review make it possible to inspect what each participant heard and whether delivery problems affected the comparison. The platform is the apparatus for studying self-similar voice; a provider benchmark or a new gaze-guidance algorithm is not the thesis's primary claim.

The empirical contribution remains pending. This draft does not report a recruited sample, condition means, significance tests, or participant quotations. A completed thesis will replace the pending

results sections with the frozen study's actual voice-condition outcomes and revise its abstract and conclusion accordingly.

1.6 Organization of the Thesis

Chapter 2 establishes the voice rationale; Chapters 3 and 4 describe the apparatus and comparison. Chapters 5–6 define measurement and analysis, and Chapter 7 organizes the pending results. The discussion and conclusion address self-similar assistance and its limits. Appendices retain instruments, data definitions, readiness decisions, and implementation details.

CHAPTER 2: LITERATURE REVIEW

The literature connects two parts of the thesis question: why a voice resembling the listener might change an interaction, and why guided search is a useful setting in which to examine that possibility. Research on voice identity and self-similarity first motivates the comparison with a participant-preferred alternative. Work on search and auditory guidance then explains how spoken assistance can have a sustained role in the task. The review closes with the sensing and contextual requirements needed to make that comparison interpretable.

2.1 Voice as a Social and Personalized Interface

Research on machine-generated speech has considered personality, source attribution, and social presence [32, 31, 25]. These perspectives motivate treating voice as more than a neutral transport for words. They do not establish that every listener will form the same impression, or that a social impression necessarily improves task performance.

For this project, personalization has two meanings. Selection of the neutral voice allows the participant to choose among existing voices in the supplied catalog. Self-similar personalization uses recorded speech to create a synthetic voice resembling that participant. Both forms may produce familiarity or liking, but they operate through different routes. Preference is selected directly in the neutral condition, whereas perceived resemblance must be checked in the self-similar condition. The manipulation is therefore not adequately described as personalized versus generic.

The bibliography also includes work on voice-assistant personality customization and similarity [38]. Personality similarity is conceptually different from acoustic self-resemblance. A listener can prefer a speaking style without hearing it as their own voice, and a close acoustic resemblance

can be experienced as uncomfortable. The planned manipulation check asks about perceived voice similarity, while the experience battery separately asks about liking, naturalness, familiarity, and uncanniness.

2.2 Self-Similar Voice and Embodied Collaboration

Self-similar voice provides the empirical motivation for this study, while auditory guidance supplies the setting in which that voice is encountered. Two particularly relevant studies examine resemblance in different roles: an avatar speaking as the player and a virtual collaborator speaking to the participant. The present study extends this question to a guiding voice without a visible agent body.

Kao et al. analyzed 657 participants in an online programming game, comparing self-similar and assigned dissimilar avatar voices while also manipulating pitch [20]. Self-similar voice was associated with more puzzles completed, longer voluntary play, and higher similarity identification, competence, relatedness, and immersion. Direct differences on the reported intrinsic-motivation subscales were not significant. These findings motivate examining several dimensions of voice experience rather than assuming that resemblance improves enjoyment or every measure of motivation.

The meaning of performance also changes across tasks. Longer voluntary play in Kao et al.'s game can indicate continued engagement; a longer fixed search trial here can indicate difficulty. Follow My Voice will therefore interpret completion time alongside workload, motivation, and reported reliance, without treating speed as a complete account of the value of self-similar voice. This is a transfer of the research question, not a prediction that the direction of every outcome will replicate.

Guo et al. used a 25-participant, within-subject design crossing appearance and voice self-similarity during virtual jigsaw collaboration [17]. Self-similar voices increased likability and believability ratings while also increasing eeriness. Voice did not significantly affect co-presence or perceived intelligence, although appearance affected those measures. These results support keeping favorable and adverse reactions separate and distinguishing voice effects from effects of a visible character. They do not establish a search-speed advantage.

Both studies compared self-similar voices with assigned alternatives; Follow My Voice instead compares the clone with a neutral voice [20, 17]. The substantive question is consequently whether resemblance to oneself offers a different experience from a voice one has already chosen. Recognition of the clone, liking of the voice, and willingness to follow its instructions need not coincide. Keeping these constructs separate makes a mixed outcome informative rather than contradictory.

The distinction between embodiment of the participant and embodiment of the agent is important. Participants wear a headset and use a tracked controller, so the interaction involves physical sensing and action. The guiding agent, however, is not represented by a co-present animated character. Measures adapted from an embodied collaborator must therefore be described as adaptations. Their use does not establish that the same factor structure or interpretation holds for a disembodied voice.

Self-voice applications in daily reminders and emotional self-voice interventions extend the range of contexts represented in the bibliography [21, 14]. They motivate asking whether participants can imagine uses for their own synthetic voice. They do not establish that navigation guidance will be more effective or that participants will experience it as inner speech. The debrief consequently begins with open questions before introducing probes about eeriness, self-talk, and possible applications.

2.3 Naturalness, Familiarity, and Uncanniness

Voice resemblance is not the only acoustic difference that listeners may notice. Synthesis can change articulation, prosody, loudness, duration, or perceived naturalness. Research on vocal uncanniness provides a reason to keep these qualities visible rather than subsume them under similarity [11]. A clone that is recognizable but difficult to understand is not equivalent to a recognizable and intelligible clone.

The experiment will use the same semantic library and provider for both conditions. This is an implementation control rather than proof of equal acoustics, because prepared clips can differ in duration and pronunciation even when their text matches. Pilot checks will therefore inspect intelligibility and comfortable playback levels across neutral and cloned voices, and the actual provider, model, voice, and library identifiers will identify the configuration each participant heard.

Custom experience items and the appealing, eerie, and familiar items selected from Guo et al. will remain separate, since combining them would imply a measurement model not established for this study. A participant may report a voice as both familiar and eerie, and preserving the individual item directions keeps that response interpretable.

2.4 Workload, Motivation, and Agent Perception

The possibility of both welcome and uncomfortable self-resemblance makes outcome selection part of the research argument. Task speed describes efficiency, while workload, motivation, and agent perception address different aspects of receiving help. The following measures retain those distinctions so that the comparison can reveal a tradeoff rather than require one voice to rank higher on every dimension.

Zagermann et al. discuss eye tracking as a means of investigating cognitive load in visual computing [43]. That motivates considering gaze-derived measures, but does not license interpreting an unvalidated ray-hit count as workload. The current confirmatory plan uses subjective workload dimensions and task performance. Pupil, fixation, and saccade analyses remain contingent on separate acquisition and event-validation work.

NASA-TLX supplies six workload dimensions [18]; the project's 0–10 response range and unweighted complete mean are documented adaptations. The Intrinsic Motivation Inventory distinguishes interest/enjoyment from related subscales such as effort, pressure, and value [34, 29, 8], and reporting the subscales separately preserves that distinction. The instrument appendix specifies the current wording and reverse keys so that scoring can be reproduced.

Agent perception measures are drawn through Guo et al., with attribution to their originating measures [17, 6, 30]. Co-presence, perceived intelligence, and anthropomorphism concern participant attributions, so a high rating does not establish sentience or decision correctness. Conversely, deterministic validity checks do not establish that a participant found the agent natural. The design therefore needs both kinds of evidence while keeping their meanings separate.

2.5 Visual Search and Attention Guidance in Extended Reality

To study a guiding voice, the task must give its instructions a meaningful role. Object search provides that role by requiring the participant to connect guidance with a location and then an intended object. The search literature helps specify this setting; it does not supply the answer to the voice comparison.

Search in a surrounding display differs from search on a single front-facing screen because relevant objects may lie outside the visible field. Turning changes what can be inspected, while eye

movements refine attention within the visible region. Work on eye and head movements across an extended field of view is therefore relevant to the task geometry [39]. In the present study, this motivates recording both head motion and gaze rather than treating one as a complete substitute for the other.

Several approaches to directing attention in immersive environments inform this design. Grogorick et al. address subtle gaze guidance, while HiveFive concerns immersion-preserving attention guidance [15, 24]. Ariza and colleagues investigate vibrotactile assistance and proximity-based multimodal feedback for selection [3, 2]. These works establish relevant design alternatives without implying that all forms of guidance are equivalent: a cue that changes visual salience, a vibration that indicates proximity, and a spoken spatial instruction impose different perceptual and interpretive demands.

RippleVision provides a recent visual counterpart to gaze-contingent spoken guidance. Kudnick et al. describe brightness ripples restricted to a cone between a point of interest and the current gaze location, with opposite brightness modulation in the two eyes [22]. This technique description is supported by the published abstract; detailed study methods and comparative outcomes remain outside the present synthesis pending full-text review. Its relevance here is conceptual: assistance can adapt to where a participant looks without specifying the target through spoken identity. It also raises a design question for voice guidance: how much assistance is useful before it becomes intrusive? That question motivates considering participant experience alongside task performance; it does not establish that visual and spoken cues have equivalent effects. Follow My Voice holds its assistance policy common across voice conditions to investigate self-similar voice, rather than comparing guidance modalities.

Binetti et al. examine visual and auditory cues for locating out-of-view objects in head-mounted augmented reality [5], and Warden et al. focus on target cue type and location in augmented-reality

search [40]. Together, these studies show why target visibility and cue availability belong in the method description. Follow My Voice does not open with a color-and-shape description of the target; instead, the sequence of directions gradually narrows the participant's search.

This hidden-target design creates an important boundary. A participant who already knows the target can continue searching without further assistance. Here, the participant must obtain enough spatial information to infer which object is intended. The voice may consequently be heard more often and may be more consequential than in a task where it supplements an already known target. That is a reason to describe the task precisely, not a reason to assume stronger voice effects in advance.

2.6 Gaze-Contingent Auditory Guidance

Auditory guidance is not a single technique. A sound can encode gaze error through loudness, indicate a location through spatial rendering, or communicate an instruction through words. Only the last of these necessarily carries the verbal and vocal identity cues central to this thesis. The following studies inform the shared search design while leaving the self-similar voice comparison open.

Losing et al. tested gaze-contingent sonification in a small desktop study with ten participants: the loudness of a flute tone varied with gaze-to-target distance, reaching silence at the target [27]. Benefits depended on the search task, with the clearest improvement in the difficult number-search condition. This is evidence for linking auditory feedback to current gaze, rather than evidence for any particular spoken phrase. The present adaptation uses short spatial instructions instead of a continuously varying tone so that participants encounter the assigned voice throughout search. Its

frequency and timing remain design choices to validate, not benefits established by the sonification study.

Kwok et al.'s Two-Step Gaze Guidance evaluated 69 participants using a three-monitor display and eye tracking, rather than a VR headset [23]. Initial spatial-audio orientation followed by finer guidance shortened search times relative to the other tested conditions; accuracy did not differ significantly across conditions. Their implementation also buffered transitions between guidance stages. The transferable idea is a staged change in the information a cue supplies: first orient toward a region, then refine within it. Follow My Voice adapts that structure to wall entry and within-wall directions; the cited study does not validate this headset, spoken wording, full-surround geometry, or transition thresholds.

Binetti et al. provide a complementary headset-based example using visual arrows and spatialized alarm sounds to locate out-of-view targets [5]. Adding sound mitigated response-time costs when visual cues and targets occupied different depth planes, but did not remove the corresponding accuracy cost. The study had no sound-only condition. Its relevance is therefore conditional support for an auditory channel when visual guidance is constrained, not proof that speech is superior to visual cues. It also motivates retaining selection errors separately from completion time in the present voice comparison.

Zhang et al.'s 96-participant AR book-search study more directly concerns verbal instructions [44]. Its audio hint specified an approximate shelf once at task onset; it was not repeated gaze-contingent speech. In the sub-study without gaze-playback feedback, hint conditions outperformed the control but did not significantly differ from one another. Combined visual/audio advantages over either modality appeared in the second task of the feedback sub-study. These results support considering task completion and workload together, while making cue timing and feedback exposure explicit. They do not establish that continuous spoken guidance or self-similar delivery will be better.

Design synthesis for the voice comparison.

The resulting design uses a common coarse-to-fine guidance policy to make the two voice conditions comparable. The agent first helps locate the relevant wall, then narrows the search within it; the participant confirms the inferred target with the right trigger. The target's initially hidden identity gives the voice an ongoing instructional role, unlike a single announcement supplementing a known target. Whether hearing that role in one's own synthetic voice changes performance, comfort, motivation, or reliance remains the empirical question.

Guidance during search, acknowledgment of a qualified gaze or controller contact, and feedback after selection are recorded separately. The researcher-facing Trial Review tool should likewise be distinguished from Zhang et al.'s participant-facing gaze-playback intervention [44]: reviewing a recording to audit delivery does not expose a participant to a learning intervention. Similarly, a success chime cannot explain an earlier change in direction within the same trial, although it may affect subsequent experience. No participant playback intervention is added to the current protocol.

The current study does not include an unguided or gaze-unaware control; both voice conditions receive the same gaze-contingent policy. Earlier project documents that considered a factorial combination of voice and gaze contingency are historical, and their interpretation cannot be transferred to this two-condition design. Any observed difference will therefore concern the assigned voice experience within the shared assistance system.

2.7 Context and Short-Term Memory in XR Agents

A related AR navigation system by Yang et al. combines Building Information Modeling with retrieval-augmented generation and compares directional arrows alone with an embodied guide accompanied by the same arrows [41]. Its comparison concerns the presentation of navigation as-

sistance, not self-similar voice. It provides a useful contrast for the present framing: the question here is how the guiding voice is experienced within a shared assistance mechanism, rather than whether a new reasoning architecture or visible agent improves navigation. The walking task, destination queries, and embodied interface in that system differ from the seated hidden-target task used here.

EmBARDiment uses gaze-informed contextual memory to support interaction with an embodied XR agent [7]. That approach motivates making recent interaction evidence available to the assistance policy, although Follow My Voice implements a narrower, bounded interaction history. This history records recent spatial progress, actual speech, and guidance transitions, and its retention limits are implementation choices rather than thresholds taken from the cited system; it does not provide long-term autobiographical memory or an inferred model of the participant's mental state. The guiding voice also has no visible agent body, so the present system does not inherit the embodiment of the cited interface.

Memory-Augmented AR Agents and GenLARP represent additional directions in the bibliography [10, 42]. The former is a conceptual framework and the latter a short system description; neither supplies a validated timing threshold for this application. User-aware virtual agents and proactive AR assistance also offer context for designing state-dependent behavior [16, 26]. These works position a design problem, rather than prove that this implementation's memory mechanism improves performance.

Short-term history is operationally useful when the same current geometry can have different meanings. A target to the left may be one the participant has not yet approached, or one they have just passed while turning right. A recently delivered cue can make another identical cue unnecessary. History distinguishes these cases only if it records reliable observations and actual playback. Holding this mechanism constant across voices means the study cannot estimate its independent effect.

Feedback specificity offers a further distinction between adaptive assistance and generic encouragement. Zimmerer et al. examined performance-specific feedback from a virtual exercise coach and found changes in selected agent-perception measures without a corresponding effect on motivation [45]. For Follow My Voice, this distinction motivates describing whether a cue reports observed progress, corrects direction, acknowledges qualified gaze or pointing, or confirms a successful selection. These functions remain common across voice conditions, so the study will compare voice configurations rather than estimate an independent effect of feedback specificity. Results from embodied exercise coaching do not establish an effect on search time in the present task.

2.8 Eye-Tracking Quality and Interpretation

Eye-tracking quality involves more than a nominal sampling frequency. The reporting guidance of Dunn et al. and the data-quality discussion of Niehorster et al. motivate reporting acquisition, validity, spatial performance, and missingness [13, 33]. The system must establish how many distinct source observations arrive, how old they are when used, and how error varies across the search space.

Validation work on the Vive Pro Eye and tools such as GazeMetrics are methodological resources [36, 37, 1]. They cannot substitute for validation on the HTC Vive Focus Vision because device model, fit, calibration, coordinate conversion, and task geometry can differ. The protocol therefore requires independent targets across heights, boundaries, distances, and head orientations, and these validation targets must be separate from targets used to estimate an offset correction.

Drewes et al. compare gaze during locomotion in virtual reality and the real world [12]. This study is seated and uses head and chair rotation rather than free walking, so locomotion findings cannot be

imported as direct estimates of its gaze behavior. Head motion will be logged because it participates in search, while claims about eye events remain bounded by the available measurements.

Salvucci and Goldberg discuss identifying fixations and saccades from eye-tracking protocols [35]. An interaction region around an object serves a different purpose. The 100-ms vicinity qualification in this system is an interface acknowledgment rule, not a validated fixation classifier. The thesis will report gaze contacts and qualified vicinity events under their own names.

2.9 Comparison of the Closest Research Strands

Table 2.1 compares the selected work by what it contributes to this design and what it leaves unresolved. It is a focused synthesis of the preceding review, not a systematic-review inventory or a ranking of systems. The literature does not supply one interchangeable baseline: spatial guidance, voice resemblance, context management, and measurement quality answer different questions.

Table 2.1: Selected research strands and their role in Follow My Voice. These are methodological connections, not evidence that the present implementation reproduces the cited effects.

Research strand	Relevance to this project	Boundary of transfer
Self-similar voice [20, 17]	Motivates the voice manipulation and both favorable and adverse experience measures.	Educational or embodied collaboration tasks do not establish a search-speed benefit here.
Auditory and staged gaze guidance [27, 23]	Supports investigating gaze-dependent, coarse-to-fine assistance.	Does not validate this headset, phrase policy, or gate durations.
AR search and task feedback [44]	Separates in-task cues from post-task feedback.	Known-target task and feedback manipulation differ from this hidden-target voice comparison.
Context-aware XR agents [7]	Motivates explicit recent context for state-dependent assistance.	Does not establish an independent benefit of this bounded memory.
Eye-tracking reporting and validation [13, 33, 1]	Informs source timing, validity, spatial validation, and gap reporting.	Reporting guidance and other-device results are not Focus Vision accuracy measurements.

The resulting contribution is a particular combination of task and comparison. The agent knows an initially unnamed target, the participant receives adaptive spatial speech, and explicit right-trigger selection remains separate from gaze. The empirical manipulation changes the voice configuration while retaining that assistance mechanism. This combination defines the scope of RQ1–RQ3; it does not imply that voice familiarity, memory, and gaze contingency have been independently manipulated.

2.10 Synthesis and Research Gap

The selected literature motivates a focused question about self-similar voice: how does guidance spoken in a synthetic version of the participant's own voice compare with guidance spoken in a neutral voice? Prior voice studies make both favorable and uncomfortable reactions plausible, while audio-guidance studies inform a shared task through which those reactions and performance can be examined.

The selected work does not address this comparison when the voice guides an initially unnamed target through a surrounding object search. The design combines a consequential use for speech with a comparator that already accommodates preference. Its contribution will be to estimate whether and how those two voice configurations differ in this setting, including outcomes that favor neither configuration consistently. The next chapter describes the shared apparatus that makes the comparison possible.

CHAPTER 3: EXPERIMENTAL PLATFORM FOR THE VOICE COMPARISON

The platform turns the voice-design question into an interaction that can be compared and reconstructed. Voice preparation establishes the two conditions, the hidden-target task gives the instructions a purpose, and a shared guidance policy responds to the participant's search. Recording ties each condition to the speech actually delivered. These functions organize the apparatus: the goal is to understand the voice experience rather than to make the decision provider an additional treatment. Figure 3.2 summarizes the pipeline; Appendix F retains its detailed state transitions, memory settings, and cancellation behavior.

3.1 Voice Preparation and Telemetry

All voice audio is pre-synthesized with ElevenLabs, which supplies both the neutral voice and the self-similar voice. To support the self-similar condition, each participant records the same standard passage in their usual speaking voice. Both conditions share the semantic phrase library, guidance policy, and playback routing, so voice condition is the intended manipulation; behavior-dependent wording, hint counts, and synthesized acoustics nevertheless need not be identical. The preparation pipeline checks wording and library readiness, but per-voice intelligibility, loudness, and acoustic comparability still require pilot verification, because automated checks do not establish perceptual equivalence. No pitch or spatial effects are applied to the self-similar voice. The system logs the provider, model, voice ID, library version, and synthesis settings.

The Trial Review application aligns gaze and controller state, guidance transitions, agent decisions, and playback events with trial geometry on a common timeline. Trial-level aggregates are stored

separately from the raw gaze, motion, and event streams. The effective gaze sampling rate must be verified from distinct source timestamps on the tested headset build rather than inferred from the nominal specification or the application polling rate. Similarly, the analysis will report ray intersections as gaze hits and will not label them as fixations unless a validated event classifier is applied.

3.2 Apparatus

The application is developed in Unity 6 (6000.3.10f1) using the Unity OpenXR Plugin (1.16.1), HTC VIVE OpenXR integration (2.5.1), and Unity XR Interaction Toolkit (3.3.1). It is deployed as a standalone Android ARM64 application on the HTC VIVE Focus Vision head-mounted display, with rendering performed locally on the headset using the Universal Render Pipeline (17.3.0). The headset supplies head-pose and eye-gaze tracking, while the tracked right-hand controller supplies pointing and trigger input. Video passthrough is available as a software option; the study described here will use the virtual-reality configuration. Network access supports cloud-based agent decisions and voice preparation; it is not used to stream rendered frames from a host computer.

3.3 Search Task

The search task will take place in a neutral virtual room. Participants will sit on a swivel chair and interact using the right controller, while virtual colored geometric objects hang on eight vertical planes arranged around the participant; Figure 3.1 shows the layout from above. Because the application holds the target identity and withholds it from the opening instructions, participants must rely on voice guidance to locate the target, which they select by pointing the controller ray at it and pressing the trigger. Stimulus density, spatial bounds, and the virtual-room and passthrough set-

tings will be verified and frozen for the study release after seated-comfort and ray-selection pilots.

For each trial, the system logs object positions, layout seed, wall assignment, and target identity.

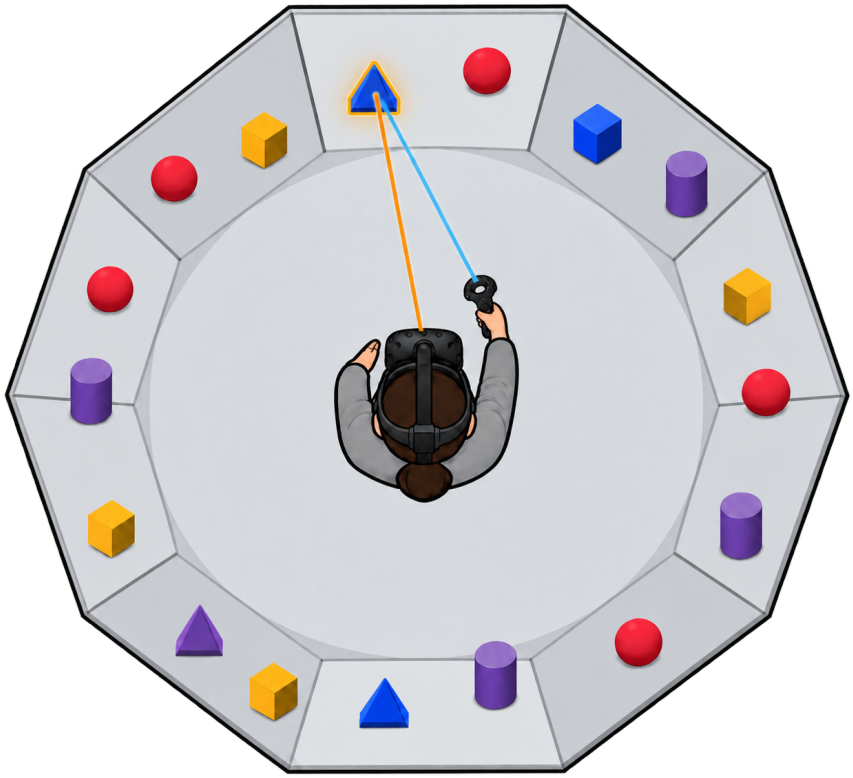


Figure 3.1: Top-down view of the search scene. The seated participant is surrounded by wall panels holding colored geometric objects. The gaze ray (orange) and controller ray (blue) converge on the highlighted target, the state in which the participant is ready to select with the trigger. The layout is illustrative; the drawn object count is schematic; object positions and target identity vary by trial. The converging rays and highlighted endpoint illustrate interaction, not an advance participant-visible target-identification cue.

3.4 Sensing and Guidance State

For repeated spoken directions to remain useful, the system needs evidence of the participant's progress. Sensing and guidance state provide that context to both voices, allowing an instruction to change from orientation toward a wall to refinement around an object.

The system continuously collects eye-gaze origin, direction, and validity; head pose; right-controller pose; and ray-object intersections. A local state machine combines these observations with the target geometry to track the participant's progress through the search: turning toward the target wall, refining within the wall, acquiring the target with gaze, pointing with the controller, and becoming ready to select. Additional states cover suspended or paused operation. Each state transition is logged with its cause.

Because guidance is only as reliable as the evidence behind it, the state machine issues new gaze-dependent guidance only when gaze data are valid and recent. Gaze ray hits, gaze-vicinity candidates, and controller hits are treated as separate observations; when gaze-vicinity regions overlap, the closest object is chosen as the vicinity candidate. A controller ray hit on the target immediately suppresses contradictory directional cues, whereas the spoken pointing acknowledgment separately requires a one-second controller hold with valid nearby gaze when speech is admitted. This suppression is intended to prevent away-directions during the qualification interval without falsely acknowledging an unqualified hold. Gaze confirmation, controller confirmation, and trigger selection are recorded as distinct events.

3.5 Agent Decisions and Playback Control

Jev, the cloud decision agent shared by both voice conditions, selects from a constrained set of options rather than generating free-form instructions. Local geometry and state rules first filter the phrase library to candidates that are valid in the current situation, and each request to Jev then contains a state snapshot, recent memory, playback status, and two to five eligible phrases. From these, Jev chooses the wording and whether to speak now or wait, while target definition and geometric reasoning remain local. Eligibility filtering and playback revalidation are intended to reduce spatially incorrect or stale guidance. Tracking error, imperfect geometry, and software faults can nonetheless produce unhelpful instructions, so recorded-sequence and headset checks remain necessary.

Because cloud responses arrive asynchronously, the participant's state may change before a decision returns. A local speech scheduler therefore revalidates each decision against the current state, evidence, pacing, and object intersections before playback, and discards decisions that are stale or conflict with the current situation. Qualified gaze and controller acknowledgments and eligible turn-progress milestones can use local handling and are played from prepared phrases without waiting for a new cloud decision, subject to the same checks. Each completed or interrupted utterance updates both the memory and the event log. Model latency, deferral, cancellation, fallback, and playback onset are logged separately to support delay attribution. When a trace lacks the required events, the cause remains unknown rather than inferred from the spoken phrase alone.

3.6 Trial Review Interface

Figure 3.3 shows the standalone Trial Review application with its bundled synthetic demonstration. The scene and inspector share a timeline with speech, gaze, events, and decision tracks. The appli-

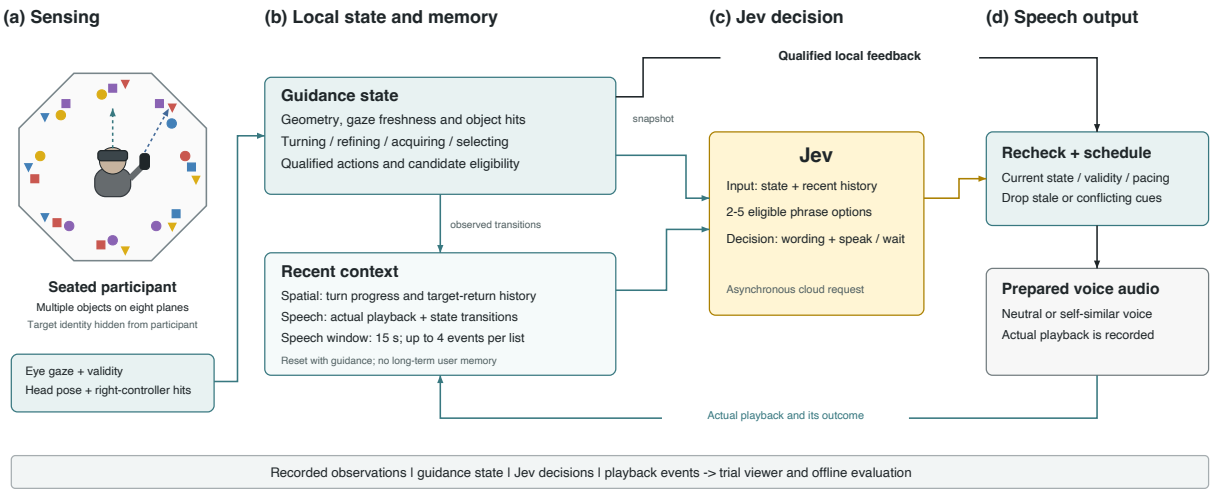


Figure 3.2: System architecture. (a) Gaze, head, and controller data are sensed. (b) A local state machine and short-term memory determine which guidance actions are valid. (c) Jev chooses wording and timing from eligible options. (d) The scheduler revalidates each cue against the current state before playback. Actual playback feeds back into memory. The scene inset is schematic and does not reflect stimulus count or any participant-visible target cue.

cation also supports receiving headset records for live review; transfer and receipt times are separate from source event times, and late arrival must not be interpreted as delayed headset guidance. This capture illustrates the review interface only; it is not a participant recording or a performance result. In particular, the demonstration explicitly marks state history as missing and highlights as reconstructed. That distinction is useful when reviewing legacy records: the viewer should expose unavailable evidence rather than imply it was measured.

3.7 Delivery Quality as an Experimental Requirement

Both voices must receive the same versioned guidance mechanism, but their delivered sequences will still depend on participant behavior and their clips may differ acoustically. The relevant control is a shared policy with documented exposure, not an assertion that every participant hears an iden-



Figure 3.3: Actual application screenshot captured for this draft using the bundled SYNTHETIC / Run 1 demonstration. The central reconstruction, right-hand inspector, and timeline provide synchronized review. Missing state, estimated rings, and reconstructed highlights are labeled by the application. This is a visualization example, not headset video or empirical study data.

tical sequence. Voice identity, library metadata, playback onset, and delivery faults must remain traceable to the corresponding condition.

A delayed or confusing cue matters here because it could alter search performance or a participant's judgment of the voice. Technical inspection therefore supports interpretation of the voice contrast. Appendix F explains how the implementation can be checked, and Appendix E preserves the inspected provider contract. Neither substitutes for the participant evidence needed to evaluate self-similar voice.

CHAPTER 4: METHOD FOR COMPARING SELF-SIMILAR AND NEUTRAL VOICES

Draft status. This chapter specifies the proposed confirmatory procedure. The implemented snapshot supports one practice, two 40-trial blocks, and final questionnaire handoff. Exact pre-cue geometry, vertical balancing, sample size, and pilot acceptance rules remain unresolved, as detailed in Appendix D.

4.1 Voice Manipulation and Comparison Rationale

The method compares two ways of personalizing assistance: a synthetic voice derived from the participant's recorded speech and a neutral voice selected by that participant from the supplied catalog. The preferred alternative makes the question relevant to a design choice a user could face, while the within-participant task provides repeated observations under both configurations. Similarity ratings will indicate whether participants heard the clone as resembling themselves; performance and experience measures will address its value as guidance.

Gaze contingency and agent decision rules are common to the two conditions. They create repeated opportunities to receive voice guidance while the target identity remains initially hidden. The experiment does not vary Jev, memory, or gaze-awareness as independent factors.

4.2 Task

Participants will perform an agent-guided object-search task while seated on a swivel chair, with multiple colored geometric objects scattered across eight surrounding vertical planes. Although the

application will know the target identity, the participant will not receive a color-and-shape description before searching. Instead, the agent will guide the participant toward the target wall and then refine the direction horizontally, vertically or diagonally. The participant will select an object using the right-controller ray and trigger; looking at an object, receiving a gaze confirmation or pointing at it will not itself complete the trial.

Each trial will begin with the participant looking at a readiness cross. Once readiness is established, an opening directional cue will play while the objects remain hidden, and object reveal will then start active search timing. A correct trigger selection will stop timing and produce success feedback, whereas an incorrect selection will be counted and search will continue. This sequence separates readiness and opening guidance from the timed search interval while retaining their timestamps for analysis.

4.3 Design and Trial Allocation

The study will use a within-participant comparison of two voice conditions: a neutral voice and a self-similar synthetic voice. Both conditions will use the same guidance policy, semantic phrase library, selection method and visual-feedback settings, although actual phrase sequences may differ because guidance responds to behavior. The comparison will therefore concern two voice configurations within an adaptive assistance system; it will not estimate the effect of gaze contingency itself. Moreover, because the neutral voice is participant-chosen while the clone is derived from the participant, preference, gender presentation and synthesis quality may differ along with self-similarity, and the contrast cannot isolate self-similarity from all of these features.

A saved, balanced randomized allocation will assign neutral-first or self-similar-first order. Each participant will complete one practice trial followed by two measured blocks of 40 trials, one per

voice. The practice trial will use the first voice in the assigned order (the neutral voice for Order A, or the self-similar voice for Order B) and will be excluded from measured outcomes. A seed will make the trial and layout schedules reproducible. Nominal horizontal-angle quotas will be identical across blocks, with trial order and left/right assignments randomized subject to those quotas. The implemented schedule, however, samples left/right signs without enforcing an equal split within each bilateral angle category, so exact four-left/four-right balance is not guaranteed; adopting that additional constraint would require an explicit protocol and implementation change before the study is frozen.

Table 4.1: Study allocation. Nominal wall angles are schedule categories, not exact object bearings or accumulated head rotation.

Parameter	Allocation
Voice conditions	Neutral voice; self-similar synthetic voice
Measured trials	40 per block; 80 per participant
Practice	One trial; excluded from outcomes
Absolute nominal wall step	Eight trials each at 0, 45, 90, 135 and 180 degrees per block
Vertical allocation	20 targets above and 20 below readiness-cross height per block (proposed; enforcement remains an implementation requirement)
Condition order	Counterbalanced neutral-first and self-similar-first
Selection	Right-controller ray and a trigger press

Objects will vary in within-plane position, so the exact starting angle and target height will be recorded in addition to the assigned wall. The vertical allocation will use target-center height relative to the readiness cross, not the participant's seated eye height. Feasible height ranges and realized quotas will be checked in pilot schedules before the study release is frozen.

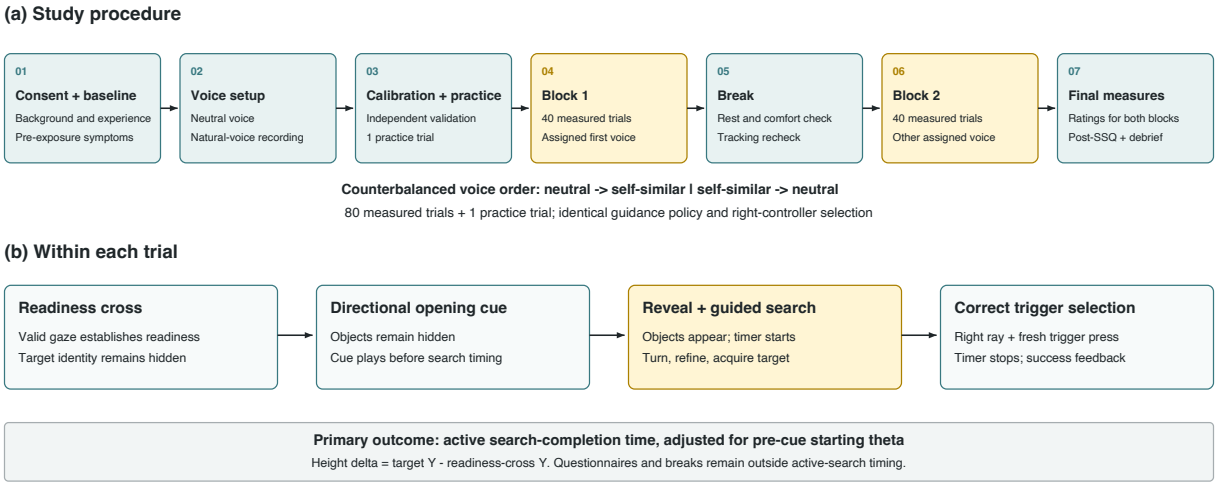


Figure 4.1: Study procedure and trial timing. Opening guidance precedes object reveal; active completion time ends at correct right-trigger selection. Questionnaires and between-trial breaks remain outside active-search timing.

Nominal angle steps are relative to the saved readiness/start wall, which the aligned schedule represents as the previous target wall center. The first measured start wall uses the practice target wall, and the completed-session check must verify the saved cross placement. This wall-center relationship is distinct from the proposed exact cross-to-target bearing.

4.4 Measures

4.4.1 Task Performance

Active completion time.

The primary outcome will be elapsed active-search time in seconds from object reveal to the correct right-trigger selection, excluding the union of recorded pause intervals overlapping that period. Chapter 5 specifies the clock equation and handling of overlapping pauses. Lower values will indicate faster completion. Readiness, opening guidance, post-selection feedback, questionnaires and between-trial breaks will not contribute to this outcome. An unfinished trial will retain its observed exposure and outcome, but will not receive a successful completion time. The full interval from opening-cue onset to selection will be retained separately to describe the complete instructed task duration.

Target contact and selection errors.

Time to first target gaze will be the active-clock time of the first valid, recorded gaze-ray intersection with the target, and time to first controller contact will use the first tracked right-controller ray–target intersection. Last target-gaze contact time will denote the last observed valid target intersection during the trial, not a fixation duration or a spoken confirmation. These event times will remain

distinct from trigger selection and from the larger gaze-vicinity region used by the guidance system. Incorrect selections will be counted as trigger selections of non-target objects. If no valid target contact is observed, its latency will remain missing rather than being set to zero, and all contact times will retain their source and sampling resolution.

4.4.2 Starting Geometry and Search Behavior

Starting angle, height and distance.

Before the opening cue, the system will record the reference origin, readiness-cross position and target position. Starting theta (θ_0) will be the signed angle about world up between the horizontal origin-to-cross and origin-to-target vectors; its magnitude $|\theta_0|$ will quantify horizontal turn demand, and its sign will preserve direction. Degenerate horizontal vectors will be marked unavailable. The primary analysis will adjust for $|\theta_0|$ rather than divide completion time by degrees, and nominal wall-angle assignment, exact pre-cue theta and head-to-target error observed after reveal will remain separate variables. The exact pre-cue snapshot is not yet implemented in the aligned study build, and existing nominal-theta fields are not substitutes for it; the snapshot must be implemented and verified before the study release is frozen.

Signed target-height difference will be $\Delta h = y_{\text{target}} - y_{\text{cross}}$, in metres; positive values will indicate above-cross targets. Euclidean target distance will be the three-dimensional distance between target center and the first valid tracked head position after reveal, with that observation's timestamp retained. This post-reveal distance will not substitute for a missing pre-cue geometry snapshot. All positions used in a calculation will share a recorded coordinate frame.

Head movement and zone coverage.

Gross head translation will sum distances between adjacent valid head positions, in metres, and gross head rotation will sum shortest quaternion angular displacements between adjacent valid orientations, in degrees. Fixed-axis component totals will be retained separately and will not be mislabeled as Euler yaw, pitch and roll. Invalid intervals, pauses and excessive sample gaps will break accumulation rather than be interpolated across, and covered time and gap counts will accompany totals. These measures describe head movement, not independently tracked chair rotation.

Distinct walls visited, wall transitions and revisits will summarize search coverage using stable valid gaze observations and a versioned zone-assignment rule. Transitions across tracking gaps will not be inferred. More motion or more visits will not by themselves establish worse strategy, since geometry and available observations also affect these totals.

Instructions and guidance timing.

The delivered active-search assistance total will count playback starts for directional guidance, gaze/pointing acknowledgments, and selection reminders, including delivered repeats; these categories will also be reported separately, as specified in Chapter 5. Requested repeats, delivered repeats and interrupted utterances will be retained separately; model requests, prefetched audio and success speech will not be counted as delivered search instructions. Opening cues and success feedback will retain separate event labels. Timing analysis will distinguish evidence qualification, decision-request latency, model deferral, scheduling delay and audio onset. Gaze confirmations and controller confirmations will be evaluated separately against their corresponding gates; the controller gate includes both its hold and the required nearby valid gaze.

4.4.3 Subjective Measures

Workload.

Six NASA-TLX dimensions—mental, physical and temporal demand, performance, effort and frustration—will be rated separately for each voice block during the final survey, after both blocks. The study will use a documented 0–10 adaptation without pairwise weighting, in which performance runs from Perfect to Failure and the other dimensions run from Very low to Very high. Individual ratings and the unweighted mean of all six completed dimensions will be retained, with higher scores indicating greater workload or poorer perceived performance. The source instrument is Hart and Staveland's NASA-TLX.

Motivation and activity experience.

The Intrinsic Motivation Inventory will contribute interest/enjoyment (7 items), effort/importance (5), pressure/tension (5), and value/usefulness (7). Responses will range from 1 = Not at all true to 7 = Very true. Published reverse keys will be applied before averaging each complete subscale. Interest/enjoyment will represent intrinsic motivation; the four subscales will not be combined into an overall IMI score. Activity wording will refer to the completed search block, with task-specific adaptations documented.

Agent perception.

Twenty-one items from Guo et al. (2024), Appendix A, Table A1 will assess co-presence (Q1–6), perceived intelligence (Q13–18), appealing/eerie/familiar impressions (Q20–22), and perceived anthropomorphism (Q37–42). "Virtual character" will be adapted to "voice agent." Responses will

range from 1 = Very strongly disagree to 7 = Very strongly agree. The three six-item constructs will have separate complete-scale means. Q20–22 will be analyzed individually, without a combined eeriness score. These attributed qualities will not be interpreted as actual agent intelligence or sentience.

Voice experience and reliance.

Twelve custom 1–7 agreement items will assess voice similarity, responsiveness, helpfulness, distraction, familiarity, trust, comfort, uncanniness, reliance, naturalness, willingness to follow and motivation to act without guidance. Voice similarity will serve as a manipulation check. Voice liking and retrospective percentage of guidance use will each use 0–100 responses with separate unable-to-answer options. These 14 numeric items will be analyzed separately; they are not a validated composite scale. Liking during voice audition will remain separate from condition-specific liking collected at the end.

Symptoms and debrief.

The 16 Simulator Sickness Questionnaire symptoms will be rated from 0 = None to 3 = Severe before headset exposure and after headset removal. Canonical total or subscale scores will require the published membership and weights. Open debrief questions will explore perceived performance, guidance experience, own-voice comfort, possible use cases and voice preference; broad experience questions will precede specific eeriness or inner-dialogue probes.

Each condition-specific set in the final survey will therefore contain 65 numeric responses (130 across both sets): six workload ratings, 24 IMI items, 21 Guo items and 14 custom ratings. Technical checks and optional text will be additional. Questionnaire observations will have participant-

by-block scope even when copied into per-trial summary files; they will not be treated as 40 independent responses. Missing answers will not be replaced with zero.

Table 4.2 summarizes the condition-specific battery. Counts refer to numeric items in one block-referenced set, not independent scales. Exact wording and reverse-key positions are preserved in Appendices A and B.

Table 4.2: Proposed final questionnaire battery for each referenced voice block. These ratings are administered after both blocks; setup voice ratings and the baseline/final SSQ are separate.

Measure	Items	Scoring and interpretation
NASA-TLX adaptation	6	Six 0–10 dimensions; complete unweighted mean. Preserve explicit bounds and adverse performance direction.
IMI	24	Separate interest/enjoyment (7), effort/importance (5), pressure/tension (5), and value/usefulness (7) means. No overall total.
Agent perception	21	Three six-item means: co-presence, intelligence, anthropomorphism. Appealing, eerie, and familiar remain three single items.
Custom voice and assistance	14	Individual ratings of voice and guidance experience. No unvalidated combined score.
Total per block reference	65	130 numeric ratings across two references, before technical checks and optional text.

4.4.4 Measurement Quality

Tracking validity, observation age, data coverage, gaps and spatial-validation error will accompany gaze-derived measures. Ray intersections and vicinity hits will not be labeled fixations, and render-frame observations will not establish native eye-tracker sampling frequency. Pupil-based workload inference and fixation/saccade outcomes will require independent validation before inclusion. Trial performance and subjective summaries will remain linked to separate raw gaze, motion and event records through participant, session, block and trial identifiers. Scale metadata will distinguish the present 0–10 workload form from any earlier exports using another range.

4.5 Research Questions

The identifiers RQ1–RQ3 and EQ1 are defined in Table 1.1; the following method-level questions retain that same scope.

The primary question will be whether self-similar and neutral voices differ in active completion time after accounting for absolute starting theta and repeated observations within participants. Secondary questions will concern workload and the four IMI subscales, while exploratory questions will concern agent perception, eeriness, voice experience, reliance, search behavior and variation in the voice contrast with starting angle. The primary voice contrast will be two-sided, so a directional benefit of self-similarity will not be assumed, and repeated trials will not be treated as independent participants.

4.6 Participants

Participants will be consenting adults able to understand the task, hear the guidance, view the scene with usual correction and safely use the seated setup and right controller. Handedness, visual correction and prior XR and voice-assistant experience will be recorded. Recruitment size and the enrollment ceiling will be determined before recruitment using power analysis for the theta-adjusted repeated-measures outcome, including plausible within-participant variation and incomplete observations.

4.7 Procedure

Before participant use, consent materials must be reconciled with ElevenLabs voice processing and Jev cloud decisions, including the data transmitted, retention, access, and deletion arrangements;

this reconciliation is a readiness requirement rather than a claim of current approval. After consent, participants will complete background questions and baseline symptom ratings. They will then choose a neutral voice, rate auditioned voices, record a standard passage in their natural voice for synthesis, and confirm comfortable, clear audio for each condition. The retained English recording passage is reproduced in Appendix A, although its match to the frozen build and typical recording duration remain to be verified. Preferred gender presentation will be chosen explicitly rather than inferred from participant gender.

The researcher will fit the headset and conduct vendor calibration followed by independent spatial validation across relevant heights, wall boundaries, distances and head orientations. If an offset correction is introduced, its fitting targets will be separate from independent validation targets; this is a conditional validation requirement, not a claim that a correction is already validated. Standardized instructions will explain that participants will rely on the agent to discover the target and use the right trigger to select it. Participants will then complete the single practice trial and the two assigned 40-trial blocks.

No questionnaire will interrupt the block sequence. Instead, during the final survey the same condition-specific questionnaire will be completed twice, once referring to Block 1 and once to Block 2, with identical item wording and order. Breaks will be offered between trials and at the block boundary, and any necessary in-trial pause will be logged immediately. Tracking will be rechecked between blocks and after headset adjustment. After headset removal, participants will first complete the final symptom check, followed by the two retrospective condition-specific sets, comparative voice preference and debrief. Participants may request a pause or stop, and symptoms or technical problems will prompt researcher assessment.

Figure 4.2 makes the distinction between administration time and reference scope explicit. Both condition-specific batteries occur in the final session; each response still refers to one measured block.

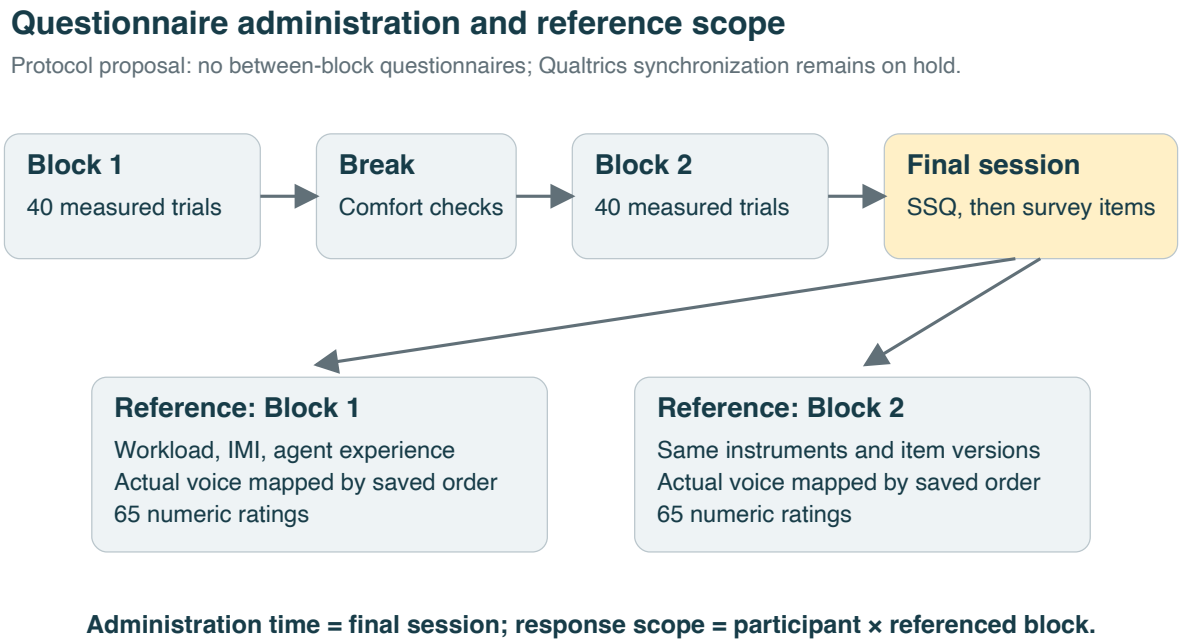


Figure 4.2: Protocol proposal for final retrospective administration. The intervening break contains comfort and tracking checks, not a block questionnaire. Each block-referenced battery contains 65 numeric ratings; the final SSQ is separate. Actual voice is recovered from the saved counterbalanced order.

The final survey will need an unambiguous reminder linking each block reference to its experienced voice. The reminder format (such as neutral labels or replayed samples) remains an open protocol decision and must be piloted without adding differential cues between conditions.

4.8 Pilot Testing and Parameter Checks

Before confirmatory recruitment, pilot sessions will verify the realized angle/height schedules, target visibility and seated comfort, readiness and selection behavior, gaze validity, audio timing and wording, and reconstruction of trial events from the saved data. Survey comprehension and completion time will also be measured. Calibration acceptance, practice proficiency, timeout and technical-stop criteria will be set from those checks and recorded with the study build and questionnaire versions. Pilot observations will be reported as such rather than presented as study outcomes or evidence for a voice-condition effect.

CHAPTER 5: MEASURING VOICE OUTCOMES AND INTERACTION EXPOSURE

The research question becomes three measurement questions: did the participant perceive the voice as self-similar, did the voice condition change search performance, and how was the assistance experienced? Answering them requires both outcome measures and an account of exposure. This chapter connects voice identity, delivered speech, search events, and questionnaire responses so that recognizing the clone is not confused with benefiting from it. Detailed gaze and timing records support the interpretation, while Appendix A retains instrument wording and scoring.

5.1 From Interaction Events to Research Variables

The recorded data serve two purposes: reconstructing what happened in a trial and estimating the study outcomes. Reconstruction requires observations and events at their original temporal resolution, whereas statistical analysis usually requires a more compact table. The project therefore keeps trial summaries separate from gaze, head/controller motion, object layouts, and event streams, treating each summary as a derived representation rather than a replacement for the underlying evidence.

Every derived value needs an identifiable source and a definition. A number called "time to gaze" is ambiguous unless it specifies the clock origin, the contact rule, validity handling, and the treatment of missing observations. The same is true of an angle: wall assignment, gaze error, starting geometry, and accumulated rotation can all be expressed in degrees, but describe different quantities. The data contract preserves these meanings in separate fields.

Figure 5.1 shows why a compact trial CSV must retain links to the underlying observations. Statistical scope and reconstruction detail impose different requirements on the exported data.

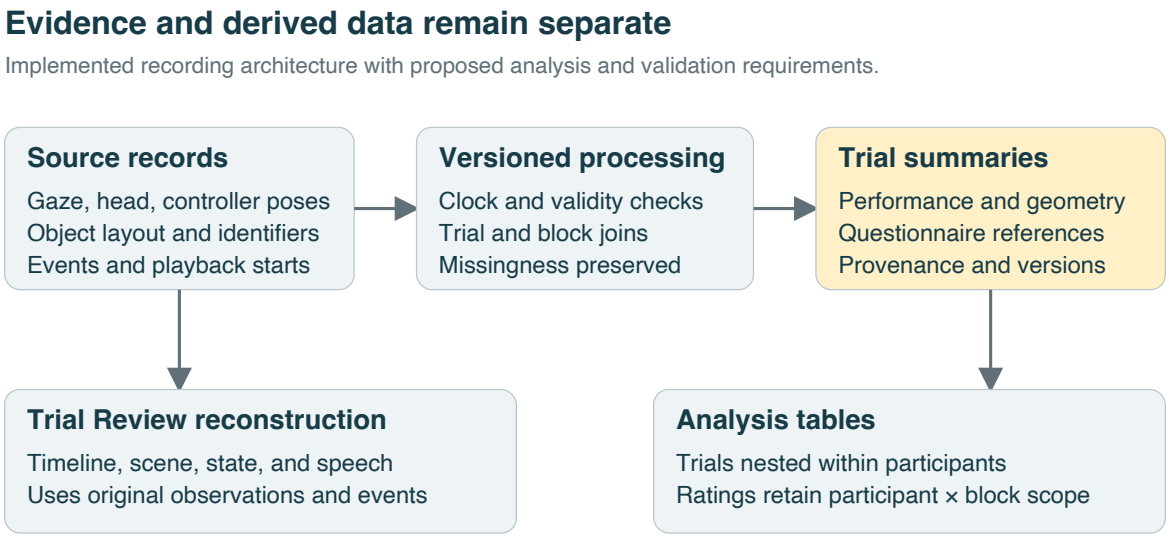


Figure 5.1: Recording and processing relationships. Source streams support reconstruction; derived summaries support analysis. Processing and scoring require versioned definitions. Repeating block-level ratings in trial rows does not create new independent observations.

5.2 Coordinate Frames and Starting Geometry

Let $\mathbf{o}$ denote the saved trial reference origin, $\mathbf{c}$ the readiness-cross center, and $\mathbf{p}$ the target center. All three positions must be expressed in the same world coordinate frame. Let $\Pi(\mathbf{v}) = (v_x, 0, v_z)$ project a vector onto the horizontal plane. Define

$$\mathbf{a} = \Pi(\mathbf{c} - \mathbf{o}), \qquad \mathbf{b} = \Pi(\mathbf{p} - \mathbf{o}). \tag{5.1}$$

For nondegenerate vectors, the exact pre-cue starting angle is

$$\theta_0 = \operatorname{atan2}(\hat{\mathbf{y}} \cdot (\mathbf{a} \times \mathbf{b}),\ \mathbf{a} \cdot \mathbf{b})\ \frac{180}{\pi}. \tag{5.2}$$

The sign convention will be verified against Unity's world-up signed-angle convention using known left and right targets. The absolute value $|\theta_0|$ describes horizontal demand from the saved cross reference. Exactly backward targets require a documented tie convention, because -180 and $+180$ identify the same horizontal bearing, although the absolute magnitude remains 180 degrees in either case.

This calculation is a required addition to the frozen study data contract. The October 20, 2026 implementation alignment audit (source register in Appendix D) found that existing signed and absolute theta fields represented nominal wall geometry relative to the previous target wall. They must not be renamed or silently interpreted as exact pre-cue measurements. A new snapshot must save origin, cross, target, timestamp, frame, and coordinate-frame identifier before the opening cue. If a horizontal vector is degenerate, the angle is unavailable; substituting zero would falsely describe an easy starting configuration.

Nominal categories remain useful for allocation: each block contains eight scheduled trials in each absolute category, $0, 45, 90, 135, 180$ degrees. Because an object can be positioned away from the center of its assigned wall, however, its exact bearing need not equal the nominal category, and balance across categories therefore does not prove exact geometric matching between voice blocks. Both assigned and realized geometry will be inspected.

The signed height difference is

$$\Delta h = p_y - c_y. \tag{5.3}$$

Positive values identify targets above the cross center and negative values identify targets below it; this reference is not interchangeable with seated eye height. The planned 20-above/20-below allocation must be checked against realized positions and a prespecified near-zero rule, because at the source snapshot, floor-to-height scattering did not enforce that balance.

The three-dimensional distance measure relies on a different observation: the first valid tracked head position $\mathbf{h}_k$ after reveal. Its value is $d = \|\mathbf{p} - \mathbf{h}_k\|_2$, in metres, and the timestamp of $\mathbf{h}_k$ must accompany the distance. Because this post-reveal measurement describes geometry at an observed head pose, it cannot recover an unrecorded pre-cue snapshot.

Figures 5.2 and 5.3 illustrate the proposed geometric contract. Neither diagram represents a recorded layout. Their purpose is to show why nominal wall assignment, exact cross-relative bearing, cross-relative height, and live gaze error require separate names and saved references.

5.3 Trial Clocks and Pause Intervals

Let t_r be object reveal, t_s correct selection, and $\mathcal{P}$ the union of valid pause intervals. Active completion time is

$$T_{\text{active}} = (t_s - t_r) - \mu\big([t_r, t_s] \cap \mathcal{P}\big)\,, \tag{5.4}$$

where μ denotes elapsed duration. Using the union prevents overlapping pause reasons from being subtracted twice. For example, an application-focus pause can overlap a researcher-requested pause. The resulting active interval should remain stopped until all relevant pause reasons are resolved.

Readiness and opening speech precede reveal and are excluded from active search. This is an intentional outcome boundary, not missing task time. The system will nonetheless retain a broader opening-cue-to-selection interval so that analyses can describe the time spent receiving the initial

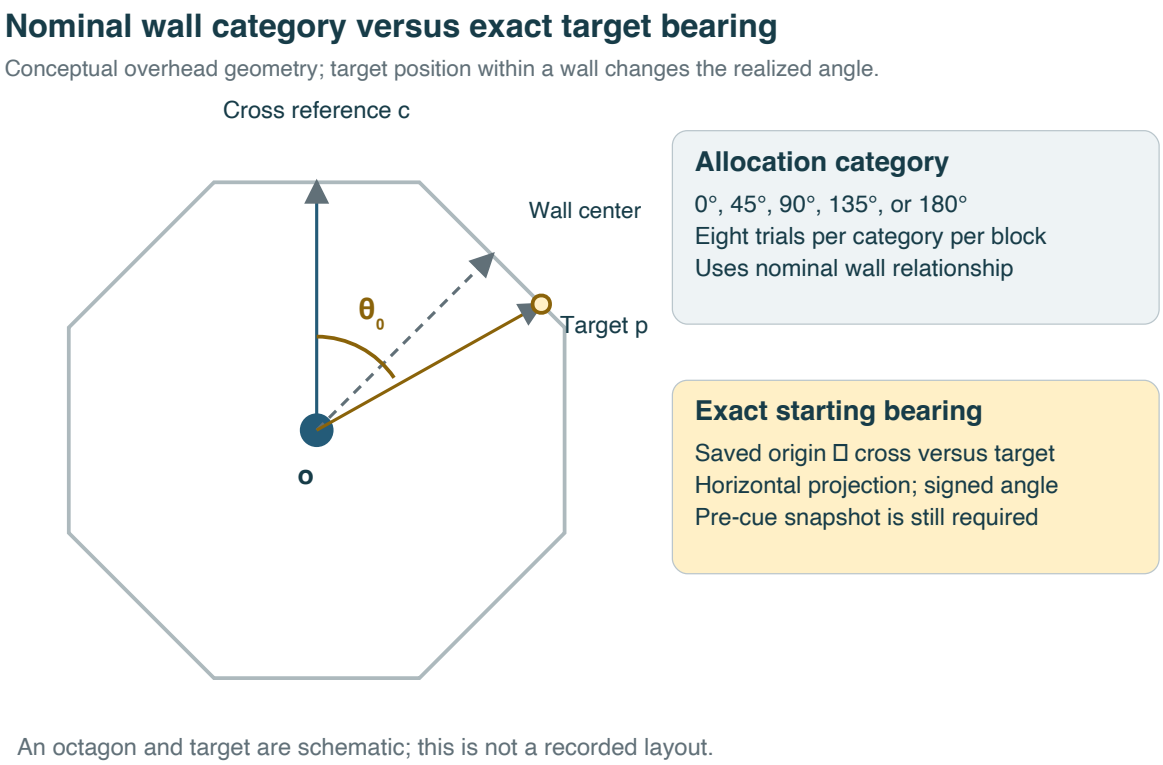


Figure 5.2: Conceptual overhead geometry. The target can be offset from its wall center, so an allocation category need not equal the realized starting bearing. The exact pre-cue snapshot remains an implementation requirement at the aligned checkpoint.

instruction; because prepared clips can have different durations across voices, this additional record remains useful even though active completion is the primary outcome.

A stopped or timed-out trial has an observed duration and an outcome reason, but it does not have a successful completion time, and the analysis must not copy its stop timestamp into a success field. Similarly, a late-arriving file in the viewer reflects a transfer delay, not necessarily a delay in the headset event, which is why source timestamps and receive timestamps are kept distinct.

Recent conversational AR research illustrates why this definition should be stated explicitly. Madritsch et al. distinguish overall task-completion time from active working time in an assistant-

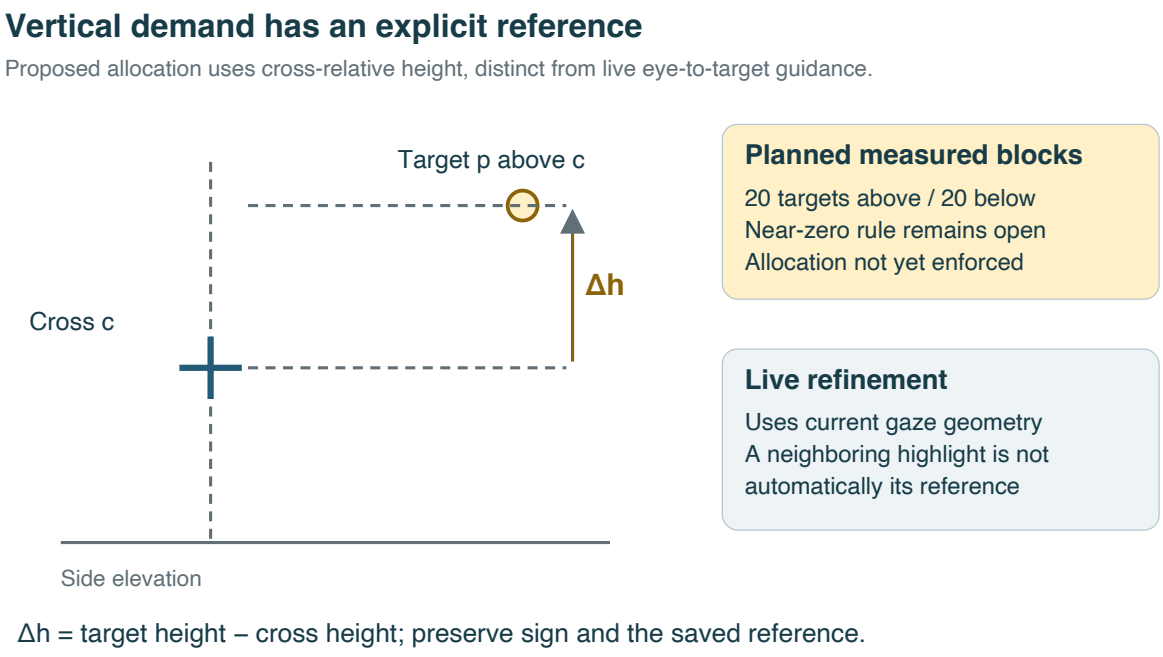


Figure 5.3: Conceptual side elevation for the proposed signed height difference. The target shown above the cross has positive height delta. The planned above/below quota and near-zero rule remain unresolved implementation and protocol checks; live vertical guidance uses current gaze geometry.

supported procedural task [28]; the present clock is not assumed to correspond to theirs. In Follow My Voice, active completion time includes ordinary waits for guidance between object reveal and correct selection, and only recorded pauses are excluded. Separate decision and playback timestamps will therefore help explain delivery delays without removing their task-level consequences from the primary outcome.

5.4 Gaze Contact, Vicinity, Pointing, and Selection

Four event families are central to interpretation: a valid eye ray can intersect the target collider; a gaze direction can satisfy the larger target-vicinity rule; the right-controller ray can intersect the

target; and a trigger press can select the target. These events can occur at different times and need not occur in that order in every recorded sample sequence.

The first gaze-contact latency uses the first valid recorded target-ray intersection after reveal. The last gaze-contact time uses the last such intersection before trial end. Neither is a fixation duration. If no contact is observed because of a blink, acquisition gap, or a path that only enters the broader vicinity, the contact field remains missing. The presence of a spoken acknowledgment does not justify filling in an unobserved ray contact.

Vicinity qualification is an interaction rule. In the documented system, a target can be acknowledged after 100 ms of valid qualifying evidence, with overlapping regions resolved to a single closest candidate. The rule trades exact collider intersection for a tolerance intended to make feedback usable, so that tolerance must be logged and piloted. The study-release tolerance remains to be frozen, and the default ring in the synthetic viewer screenshot is not a validated study setting. A very large region can produce premature acknowledgment even when its timing is correct.

Controller pointing is a distinct input observation, but its spoken acknowledgment is not independent of gaze: the documented rule requires a qualified one-second controller hold with valid gaze in the target vicinity when speech is admitted. Before the hold completes, a valid target hit already suppresses contradictory away-directions; this suppression prevents a direction such as “go back down” from being delivered while the controller is already on the target, without falsely saying the hold has qualified. Correct trigger selection is the only event that successfully completes the search; a stop or timeout has a different outcome.

Figure 5.4 contrasts the three feedback and completion events. The example utterances communicate their meanings and are not asserted to be verbatim clips in every voice library.

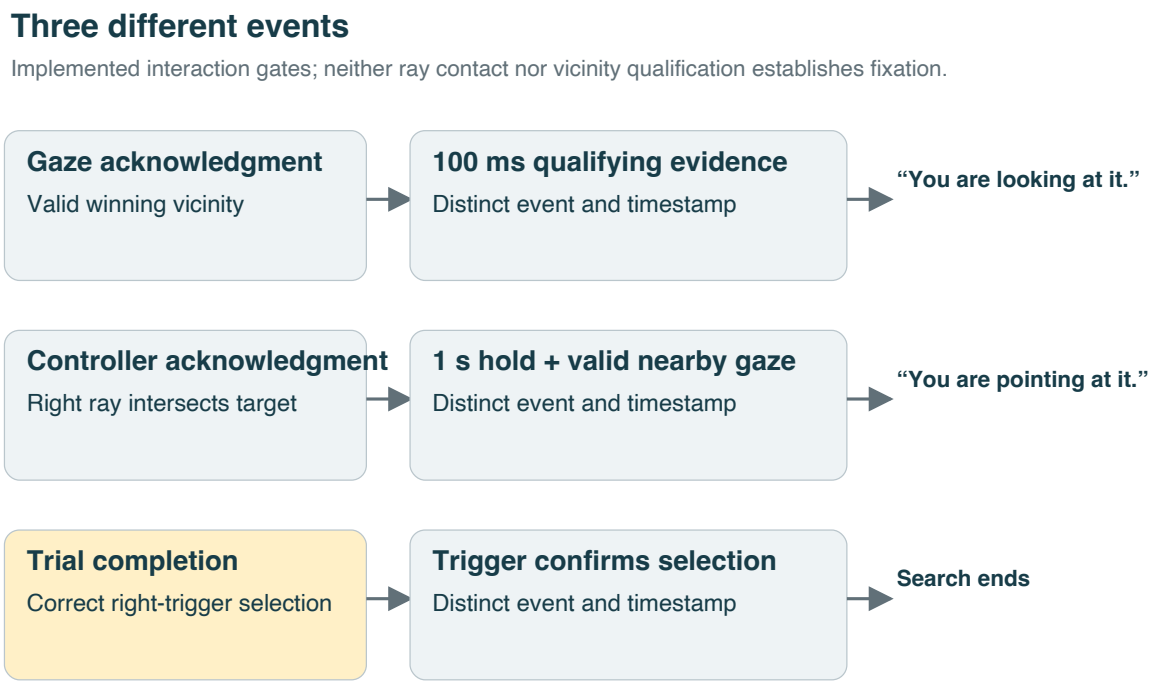


Figure 5.4: Implemented interaction distinctions. Gaze acknowledgment uses a 100-ms qualifying vicinity gate; pointing acknowledgment uses a one-second controller hold with required valid nearby gaze. A correct trigger selection completes the trial independently of whether either acknowledgment was spoken. These are interaction gates, not validated fixation measures.

5.5 Acquisition Rate, Validity, and Gaps

A render loop can query an eye-tracking API frequently without receiving a new hardware observation each time, so the effective acquisition rate must be estimated from distinct source observations. If n unique observations span source times t_1 through t_n, a basic rate summary is $(n-1)/(t_n - t_1)$. Because a mean alone can conceal bursty delivery, the distribution of successive intervals, repeated timestamps, invalid observations, and long gaps must accompany that average.

Application timestamps describe when data were handled, whereas source timestamps describe when the tracking observation was produced, if the API exposes them reliably. A clock-domain mapping must therefore be established before age or latency is computed across these clocks. If unique source timing is unavailable, the report will state that limitation rather than infer 120-Hz binocular delivery from polling or headset specifications.

Spatial validation will compare gaze direction with known independent targets. Angular error will be computed from normalized gaze and expected target directions, with invalid observations excluded from the error distribution but retained in availability summaries. Precision and bias should be considered separately, since a stable offset and a noisy signal can produce different guidance failures even when their average angular error is similar. The validation procedure must also describe target exposure, head pose, corrective lenses, and headset adjustments.

5.6 Motion, Coverage, and Instructions

Gross translation will sum Euclidean differences between adjacent valid head positions, and gross rotation will sum shortest quaternion angular differences; invalid intervals, excessive gaps, and pauses break the accumulation. Gross movement is a total, not a rate. If a separate movement-rate outcome is derived, its denominator will be observed covered time rather than the entire nominal session, because otherwise two runs with different data loss could produce misleadingly different movement rates.

Fixed-axis components derived from rotation vectors are not Euler yaw, pitch, and roll, and their labels will retain the actual calculation. Head motion also does not directly measure swivel-chair rotation, because the head can move relative to the torso and chair. Likewise, the seated avatar

drawn at the scene center in Trial Review is a visualization convention, not evidence that measured head translation was zero.

Instruction counts will use actual software playback starts; a requested cue, a rejected model answer, and a prefetched clip are not delivered instructions. Opening cues, search guidance, acknowledgments, reminders, and success speech will have separate categories. The planned total of delivered active-search assistance includes directional guidance, gaze/pointing acknowledgments, and selection reminders, with each category also reported separately, while opening and success speech remain outside that total. Interrupted speech counts as started speech but retains its completion status, so it should not be treated as if the participant heard every word.

5.7 Sequence-Level Latency

For a qualified event, let t_q denote the earliest supported qualification time and t_p the actual software playback start of the corresponding acknowledgment. The primary engineering delay is

$$L_{\text{software}} = t_p - t_q. \tag{5.5}$$

This is not acoustic latency at the participant's ear, because audio-device buffering and the physical sound path require a separate measurement. Gaze and controller qualification times must also use their own definitions, since comparing a controller acknowledgment against first gaze contact would inflate or misattribute its delay.

Repeated requests about one continuously qualified event form a single episode, so a sequence can contain several Jev waits without creating several independent missed confirmations. An episode ends with playback, disqualification, trial termination, or a defined reset, and the evaluation will report how those endpoints were assigned, marking incomplete evidence as unknown.

Where logs support attribution, waiting will be divided into model response, explicit model deferral, scheduling, existing speech, and audio readiness. These intervals may overlap. They should not be added mechanically unless represented as a nonoverlapping partition of the critical path. The full event timeline remains the authority when aggregate attribution is ambiguous.

Figure 5.5 separates intentional evidence qualification from subsequent delivery delay. A model response can be timely while playback is delayed elsewhere, and overlapping waits must not be double-counted.

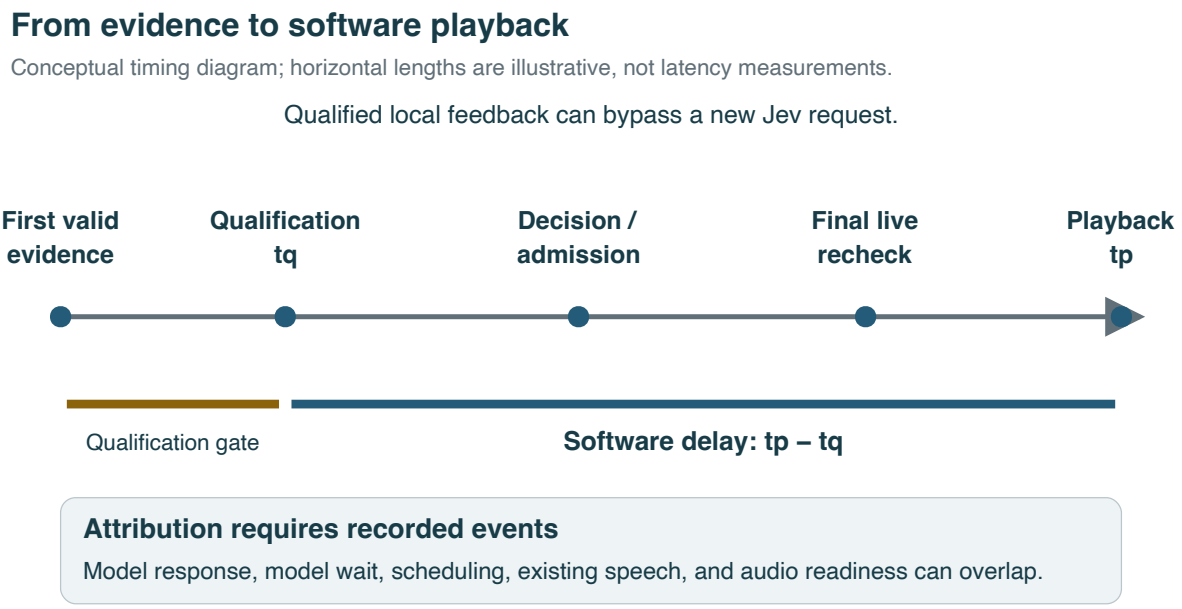


Figure 5.5: Conceptual event timeline, not a measured trace. Software delivery delay begins at supported qualification and ends at actual software playback start. The preceding qualification gate and the unmeasured acoustic path are separate quantities. Qualified local feedback can avoid a new cloud request.

Table 5.1 keeps the principal timing variables separate. In particular, the onset of an audio clip is a software event; neither a candidate phrase nor a successful provider response proves that the participant heard it.

Table 5.1: Timing contract for planned analysis and engineering evaluation. All cross-clock comparisons require an established mapping; unavailable evidence is not assigned a zero duration.

Quantity	Endpoints	Interpretation limit
Active completion	Reveal to correct trigger, less union of pauses	Excludes readiness and opening cue; unfinished trials remain unfinished.
First gaze contact	Reveal to first valid target-ray hit on active clock	Intersection, not fixation or vicinity acknowledgment.
First pointing contact	Reveal to first tracked right-ray hit on active clock	Not the one-second hold or trigger selection.
Jev response latency	Request dispatch to response receipt	A response can be deferred, canceled, or stale.
Confirmation delivery	Qualified evidence to matching actual software playback	Separate gaze and pointing episodes; not acoustic latency.
Live transfer lag	Source event to viewer receipt, if clocks are mapped	Describes transport, not headset decision delay.

5.8 Questionnaire Joins and Versioning

Final questionnaire administration is separate from the block to which a response refers. A response collected after both blocks will therefore retain both its final-administration timing and a Block 1 or Block 2 reference, and the actual voice identifier must be joined through the saved order rather than inferred from block number. Practice is excluded from measured block outcomes even though it uses the first voice.

A convenience export may repeat a block-level workload score on every trial row. That does not create 40 independent workload observations. The analytical dataset will identify participant-by-block scope and collapse duplicate joins when estimating questionnaire effects. Raw responses, score direction, scale bounds, item identifiers, instrument version, and missingness will remain available.

The current NASA-TLX adaptation uses 0–10 ratings, whereas older development exports used 0–100, so automatic scoring must require explicit range metadata. A separately named normalized score can support comparison, but raw values must remain unchanged, and unknown legacy performance anchors cannot be repaired by guessing their direction. Complete-subscale rules for IMI and agent-perception measures will likewise be implemented and tested before confirmatory analysis.

CHAPTER 6: PLANNED STATISTICAL ANALYSIS

The analysis asks whether the two voice configurations differ in efficiency and experience, while allowing those answers to diverge. Active completion time is the primary outcome, workload and motivation are secondary, and agent experience, reliance, and debrief responses are exploratory. The repeated-measures design connects those outcomes to the same participants under both voices. Guidance diagnostics describe exposure and technical context. Because Jev and gaze contingency are shared by both conditions, the statistical comparison concerns voice configuration rather than their independent value.

6.1 Primary Contrast

The primary estimand will compare active completion time under the self-similar and neutral voices, standardized to the same starting-difficulty distribution. A trial-level mixed-effects analysis of log completion time will account for participant clustering, voice condition, period/order, trial progression and absolute pre-cue starting theta. The analysis will report the voice time ratio and uncertainty using a two-sided contrast.

Nonlinearity in starting difficulty will be represented by a prespecified low-complexity function of theta, while nominal-angle strata will support balance plots and a categorical sensitivity analysis. Zero-degree trials will remain in the analysis, and completion time will not be divided by angle. Before confirmatory recruitment, simulation will be used to finalize the model's random-effects structure, angle function and smallest meaningful effect. The participant count will be justified under that same model rather than by treating repeated trials as independent participants.

Figure 6.1 illustrates the planned units of observation. Trial-level measurements are nested within participants, while retrospective ratings remain participant-by-block observations.

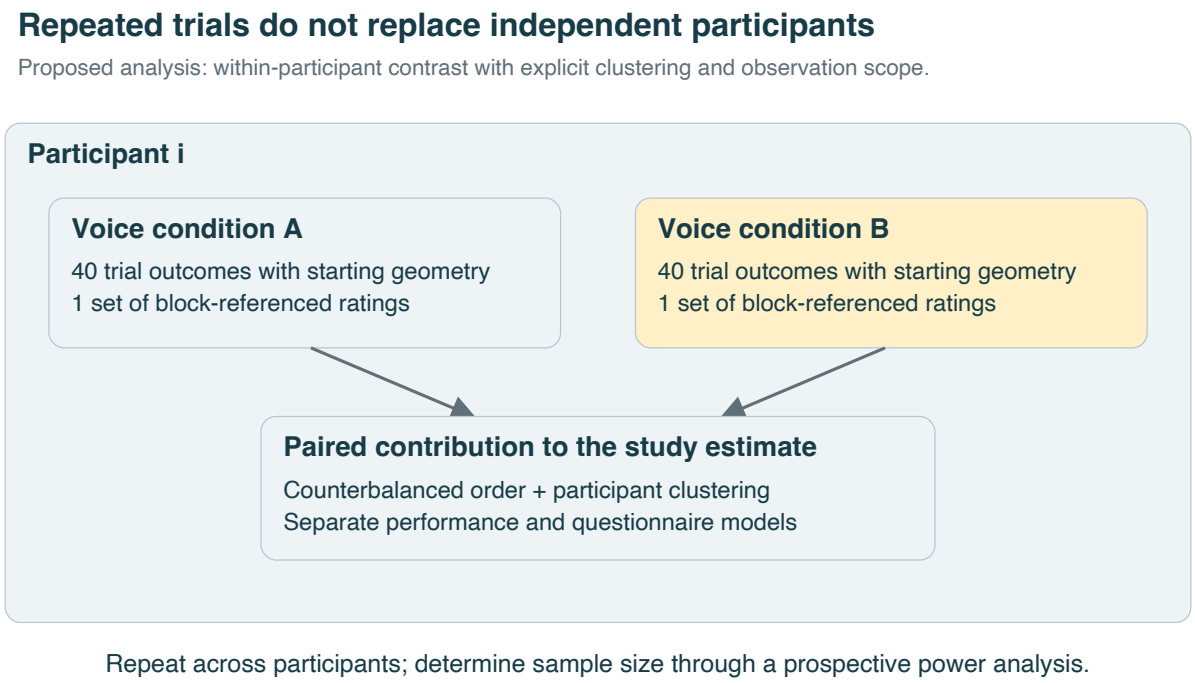


Figure 6.1: Proposed analysis scope. Each participant contributes repeated trial outcomes and two block-referenced questionnaire sets. Counterbalanced order and participant clustering remain part of the analysis plan; additional trials do not replace a justified participant sample size.

Table 6.1 summarizes the proposed outcome hierarchy. It is an analysis plan rather than a results table; final contrasts, missing-data handling, and multiplicity families require preregistration.

Table 6.1: Outcome hierarchy and observation scope. Engineering metrics diagnose the guidance system and do not replace participant outcomes.

Role	Unit	Planned interpretation
Primary performance	Trial within participant	Voice contrast in active time, adjusted for starting geometry and design factors.
Secondary experience	Participant by referenced block	Workload and separate IMI subscales, retaining paired scope.
Exploratory perception	Participant by referenced block	Agent constructs, individual custom items, and voice-similarity manipulation check.
Safety	Participant over full session	Baseline-to-final symptoms and stops; no separate voice-specific SSQ effect.
Engineering quality	Qualified event episode / request	Delivery delays, errors, repetition, and provider reliability with distinct denominators.
Qualitative experience	Participant response / coded excerpt	Debrief themes with an audit trail, not causal performance estimates.

6.2 A Candidate Primary Model

A candidate model makes the intended estimand explicit while leaving the final specification to preregistration. For participant i, period b, and trial j, let $Y_{ibj} = \log T_{ibj}$ for successfully completed active-search intervals. A starting specification is

$$Y_{ibj} = \beta_0 + \beta_V V_{ib} + \beta_B B_b + \beta_O O_i \tag{6.1}$$

$$+ f(|\theta_{0,ibj}|) + g(j) + u_{0i} + u_{Vi} V_{ib} + \epsilon_{ibj}. \tag{6.2}$$

Here V indicates self-similar rather than neutral voice, B identifies the second period, and O identifies assigned voice order. The functions f and g represent prespecified low-complexity effects of exact starting angle and trial progression. In this candidate specification, j indexes position within a block (1–40); the period term represents the between-block change. The exact pre-cue angle requires the snapshot described in Chapter 5, which remains an implementation gap. This

model cannot be fitted as specified using nominal wall angles relabeled as exact measurements. Participant intercepts and, if supported by the design and model stability, participant voice slopes represent clustering and variation in within-participant contrasts. With only one block per voice, the slope variation can also absorb block-specific session effects; it is not a pure measure of stable individual voice responsiveness. The final random-effects structure is an open decision, informed by the experimental design and simulation rather than selected to obtain significance [4].

Under this model, the exponentiated voice coefficient, $\exp(\beta_V)$, is a conditional multiplicative time contrast and will not automatically equal a marginal arithmetic mean ratio. The primary estimand therefore remains the marginal contrast standardized over the prespecified common starting-difficulty distribution; the conditional coefficient will be reported separately as a model parameter rather than substituted for that estimand. Predictions on the original seconds scale require an explicit retransformation procedure and uncertainty estimate. No fitted coefficient is available in this first pass.

A two-period design supports adjustment for period and sequence, but cannot identify every possible asymmetric carryover mechanism independently. For example, using a voice during practice could affect familiarity in its first block, and the second block could benefit from task learning. The report will describe this limitation rather than claim that adding an order term removes all carryover.

6.3 Sample Size and Precision Planning

Forty trials per block increases within-participant information, but does not turn one participant into 80 independent people. A power analysis must reflect the planned trial-level model, between-participant heterogeneity, within-participant correlation, geometry distribution, incomplete trials,

and the effect size worth detecting. An old calculation for a different number of conditions cannot justify this design.

The proposed simulation workflow will generate participants with both voice blocks and the intended nominal-angle quotas, varying plausible residual variation, participant intercept variation, voice-slope variation, and missingness. Each simulated dataset will be fitted with the same model intended for the final analysis, and convergence, bias, interval coverage, and power or precision will be summarized across candidate enrollment sizes. The chosen smallest meaningful time ratio and its supporting assumptions will be documented before recruitment. Synthetic pilot values used to test the workflow will be labeled as such and will not be presented as empirical variance estimates.

A retained pilot can inform variance assumptions only under a documented rule about whether those observations enter the confirmatory sample. Reusing pilot outcomes after extensive policy tuning risks treating development data as untouched confirmation. The study will preserve a build and protocol identifier so that a pilot collected under a different guidance policy can be separated from the frozen release.

6.4 Secondary Outcomes and Multiplicity

Condition-specific ratings will be collected retrospectively, only after both blocks, with actual survey timestamps retained; each rating set nevertheless remains referenced to a participant-by-block observation. Because these are not immediate post-block measurements, recall and recency effects will be considered when interpreting condition differences. Individual workload dimensions and complete unweighted workload means will be summarized by participant and block. IMI subscales and the three six-item agent-perception constructs will remain separate scores, and the additional Guo Q20–22 appealing, eerie and familiar ratings will be analyzed individually rather than com-

bined into an eeriness score. Voice similarity will serve as a manipulation check, while custom experience ratings and self-reported guidance use will also be analyzed individually. Multiplicity families and secondary contrasts will be specified before confirmatory analysis.

The analysis will retain paired observations and account for period where appropriate. The choice between ordinal-item and approximately continuous scale models will be made before confirmatory analysis and justified for each measure. Reporting only the items with favorable differences would distort the battery's intended scope.

Multiplicity families and adjustment rules remain to be finalized. Holm's procedure is available for a prespecified family of confirmatory secondary comparisons [19]; it is not a substitute for deciding which outcomes belong in that family. Exploratory agent-perception and custom-item results will be labeled, with effect estimates and uncertainty rather than an indiscriminate list of significance claims.

Voice similarity is a manipulation check. Failure to recognize the clone is informative about the manipulation and will not automatically remove a participant. Likewise, low liking is an outcome, not a reason to exclude an inconvenient response. Sensitivity analyses may investigate synthesis quality or recognition, but conditioning on post-assignment experience changes the interpretation and must be stated.

6.5 Data Screening and Missingness

Data screening will first establish structural integrity: unique participant/session/block/trial keys, consistent voice assignments, chronological reveal and selection events, and required geometry provenance. It will then assess observation coverage and protocol deviations. This order avoids

excluding a slow trial merely because it looks unusual while overlooking a malformed timestamp or an incorrect condition join.

Tracking loss can affect guidance and measurement simultaneously, though not always in the same way. A trial with sparse gaze coverage may have valid selection timing yet unreliable gaze-contact latency, so exclusion should be measure-specific where justified. A missing gaze contact does not automatically invalidate completion time, and a complete motion stream does not repair missing voice playback events. Each analysis will therefore report its own denominator and reasons for missingness.

Stopping, timeout, withdrawal, and technical failure require a prespecified treatment. A complete-case time model estimates completion times among observed successful trials under its assumptions. It may not describe all assigned trials when incomplete outcomes differ by voice. The final plan must define a complementary completion-probability or time-to-event analysis if incompletion is consequential. A timeout threshold has not been frozen in the source protocol and is not invented here.

Raw attempts, incorrect selections, pause intervals, timeouts, technical failures and withdrawals will all be retained, and stopped trials will not be coded as successful completion times. Timeout and censoring rules, together with the missing-data estimand, will be specified before collection so that slow trials are not removed solely for being slow. Missing responses will not be replaced with zero; questionnaire scores will instead follow the complete-subscale rule, with item-level missingness reported. Disliking or failing to recognize a cloned voice will not by itself exclude a participant.

6.6 Behavior as Outcome Rather Than Automatic Adjustment

Instruction count, gaze path, and controller behavior can be consequences of voice condition. Including them automatically as primary-model covariates could remove part of the effect the study intends to estimate. The primary comparison therefore adjusts for pre-cue difficulty and design variables. Voice-by-theta variation, selected neutral voice, acoustic quality, guidance timing, and search behavior will enter exploratory or sensitivity analyses with their assumptions reported.

For example, slower search can produce more opportunities for guidance, and more guidance can itself affect search time. A correlation between instruction count and duration does not establish that speech caused delay. Timeline analysis can identify concrete scheduling failures, but observational associations among adaptive behaviors require more cautious interpretation.

6.7 Guidance Delivery Evaluation

Engineering analysis will distinguish event qualification, decision latency, model deferral, scheduling delay and actual playback, and it will evaluate gaze and controller confirmation separately. Because a cue can be judged only in context, sequence-level review will include what happened before and after it. Automatic checks and ratings produced by an offline language-model evaluator (Appendix F) will remain advisory engineering evidence rather than participant perception scores.

6.8 Qualitative Analysis and Mixed Evidence

The debrief will begin with broad experience questions, followed by balanced probes about voice preference, perceived performance, comfort, and potential uses. The coding process will record the

initial codebook, revisions, coding decisions, and independent review of a subset. Contradictory or minority accounts will be retained rather than forced into one favorable theme.

Reported reliance and observed behavior will be considered together without treating either as a perfect measure of the other. A participant may say they relied on the agent while rapidly anticipating several cues, or may remember one frustrating interruption despite mostly timely guidance. Such differences can guide interpretation but do not establish dishonesty or measurement failure. The final report will separate quotations, coder interpretation, and behavioral evidence.

Perceived performance, voice comfort, reliance, and possible own-voice use cases will be interpreted alongside, but not substituted for, measured task performance.

6.9 Safety Outcomes and Manipulation Checks

Baseline and final Simulator Sickness Questionnaire responses will describe symptom burden and pre-to-post change across the full exposure, with canonical scores calculated only from the intended item anchors and scoring rules. Because there is no between-block administration, these measurements do not provide separate voice-block sickness effects. Safety stops, discomfort reports, and incomplete exposure will be reported alongside symptom change. This draft does not invent a numerical exclusion or stopping cutoff, but the operational stopping procedure must be frozen before participant collection.

The similarity manipulation check will report the paired difference between self-similar and neutral-voice ratings with uncertainty. A group-level decision rule and any practically meaningful difference remain open preregistration decisions. The study will not classify an individual as invalid merely because the clone is not recognized. Because equivalence testing is not currently specified, a nonsignificant voice contrast will not be interpreted as equivalence.

CHAPTER 7: VOICE-COMPARISON RESULTS FRAMEWORK AND STUDY READINESS

The completed results chapter will follow the same question introduced at the start of the thesis: what changes when guidance sounds like the participant rather than a voice they prefer? It will establish the delivered comparison, report search efficiency, and then examine workload, motivation, and exploratory experience. No confirmatory participant outcomes are available in this draft. The reporting framework below keeps those unanswered questions visible and separates them from the engineering evidence available for the apparatus.

7.1 Participant Results to Be Added

The completed empirical chapter will first report participant flow, exclusions, incomplete sessions, and data coverage by condition and order. It will then describe realized starting geometry and the similarity manipulation check before presenting the primary voice contrast. This order allows readers to assess whether the intended comparison was delivered and measured.

The primary result will include the fitted time contrast, its uncertainty, the analyzed participant and trial counts, and model diagnostics. Secondary sections will then report workload and IMI subscales at participant-by-block level, followed by exploratory sections on agent perception, voice experience, guidance behavior, and debrief themes. Tables will identify missing observations rather than silently change denominators.

Table 7.1: Planned empirical results, not observed findings.

Result family	Required evidence	Present status
Participant flow	Consented, started, completed, stopped; reasons	Not available
Manipulation check	Similarity by voice with paired scope	Not available
Primary performance	Adjusted time contrast and uncertainty	Not estimated
Workload and IMI	Versioned raw items and complete subscales	Not collected for this report
Agent experience	Individual and construct-specific ratings	Not collected for this report
Guidance quality	Sequence-level playback and evidence	Evaluation design specified
Debrief themes	Recorded responses and coding audit	Not available

7.1.1 Results Organized by Research Question

The completed results chapter will report the voice-similarity manipulation check before the outcome contrasts. RQ1 will use the marginal time contrast standardized over a common starting-difficulty distribution, as defined in Chapter 6; this is not a standardized effect-size label. Its uncertainty, eligible participant/trial counts, and model diagnostics will accompany the estimate, using the analysis plan finalized before confirmatory recruitment. RQ2 will report the paired workload and separate IMI outcomes with multiplicity families to be finalized in the analysis plan. RQ3 will report exploratory agent-perception and custom ratings alongside debrief themes and valid search measures. A missing or inconclusive contrast will be identified as such rather than omitted.

EQ1 will be reported separately through sequence coverage, qualification-to-playback delay, provider reliability, and categorized faults. A clean engineering test does not answer a voice-effect research question. Conversely, a voice-condition difference does not by itself show that the guid-

ance mechanism was correct. This separation follows the evidence map in Table 1.1 and prevents a successful build or a favorable judge rating from becoming an empirical conclusion.

7.2 Evidence Available for This Draft

This chapter separates existing engineering records from participant results that have not yet been collected. The October 20 alignment audit documents software checks, analysis tests, and an Android build/install/launch. These records support specific implementation claims, but they do not establish a treatment effect, a naturalness rating, eye-tracker accuracy, or acoustic response latency.

The audit reports passing schedule and voice-isolation checks, including one practice trial in the first assigned voice, two measured blocks, and nominal-angle quotas, and it records 38 trial-export tests and seven workload scoring/plot tests. These counts describe the audit's test run rather than a newly executed suite for this manuscript. The verification record should therefore accompany any later thesis statement so that the tested revision remains identifiable.

7.3 Guidance Quality Supporting the Voice Contrast

The completed report will describe gaze availability, actual speech exposure, delayed acknowledgments, interruptions, and unhelpful cues by voice condition and session. This account will help identify whether a condition was affected by a technical delivery problem. It will not automatically adjust away behavior-dependent hint counts or waiting time: these arise after voice assignment, may themselves depend on voice condition, and are not interchangeable with baseline difficulty, as discussed in Chapter 6.

Table 7.2: Evidence status at the documented implementation checkpoint.

Evidence	Supported conclusion	Boundary
Schedule and voice checks	One practice and nominal 40-by-two allocation pass fixtures	Does not verify wearer completion of a full session
Trial export and TLX tests	Declared scales and retrospective joins pass tested cases	Not validation of all questionnaires
Unity integration checks	Listed state and interaction fixtures pass	Not proof against all timing races
Android build/install/launch	Application built, installed, and started	Not a participant acceptance session
Exact pre-cue theta	Required definition documented	Snapshot implementation remains a gap
Vertical allocation	Cross-relative definition documented	20/20 enforcement remains a gap
Voice-condition outcomes	No confirmatory dataset reported	No performance or preference conclusion

Gaze acknowledgment, controller acknowledgment, and correct trigger selection remain separate events. Their sequence records can establish what feedback was delivered, while participant ratings and debriefing address how that feedback was experienced. An advisory model judge cannot supply the latter evidence. Detailed policy pilots, episode definitions, failure categories, and the investigation of two-second delays are retained in Appendix F; their purpose is to qualify the apparatus before interpreting a self-similar voice effect.

CHAPTER 8: DISCUSSION AND LIMITATIONS

8.1 What the Study Can Explain

The practical question is how to choose a voice for immersive assistance. Follow My Voice addresses one part of that decision: whether a synthetic voice resembling the listener changes search efficiency or experience relative to an already preferred alternative. This chapter sets out how the planned findings could answer that question, connect with prior voice research, and inform comparable applications. Because participant results remain pending, the interpretations are conditional rather than claims that either voice has proved better.

The discussion will distinguish three levels of evidence. RQ1 concerns measured efficiency; RQ2 concerns workload and motivation; RQ3 explores how participants describe and evaluate the voices and their reliance on guidance. Agreement among these outcomes could strengthen a design interpretation, while disagreement could reveal a useful tradeoff. Recorded speech and interaction provide the technical context for each interpretation, but cannot establish why a participant welcomed or ignored the voice.

8.2 Interpreting Efficiency and Experience Together

If self-similar voice produces faster active completion with acceptable uncertainty, the interpretation will still depend on the manipulation check and guidance quality. A difference accompanied by systematic intelligibility or delivery problems in the neutral condition could reflect those problems rather than anything specific to self-similarity. If both conditions have comparable delivery quality and participants recognize the clone, the result would more directly support a voice-condition

effect in this task, although it would still not separate resemblance from other differences between the chosen and synthesized voices.

If the neutral voice produces faster completion, that would be consistent with a selected external voice being easier to follow, or with intelligibility or delivery problems in the cloned condition; delivery records and acoustic-quality reporting could help distinguish these readings, but the study would not establish the mechanism without further evidence. Liking, naturalness, and reliance ratings may help describe the experience. They should not be treated as proven mediators solely because they correlate with performance.

If completion times are similar, the width of the uncertainty interval becomes central. A wide interval would leave practically important differences unresolved, whereas a narrow interval near the null could constrain the size of a plausible effect under the tested conditions. A nonsignificant test alone will not establish equivalence. Equivalence testing is not part of the current analysis specification, and adding it would require a meaningful bound and procedure defined before inspecting the result.

Experience and performance may diverge. Participants may prefer their cloned voice while searching no faster, or may find it eerie while completing efficiently. Such outcomes would be substantively useful because they distinguish usability, affect, and speed. The battery is designed to retain these differences rather than force a single ranking.

Prior voice studies help explain why such a tradeoff would be informative. Kao et al. examine an avatar voice in a setting where continued voluntary play can reflect engagement, whereas Guo et al. report favorable impressions alongside eeriness in virtual collaboration [20, 17]. Here, a voice instructs the listener through a fixed search task. Faster completion therefore addresses efficiency, while enjoyment, discomfort, and debrief responses address the experience of following it. Dif-

ferences across these studies would need to be interpreted in relation to the voice's role and the comparison condition, rather than treated automatically as agreement or contradiction.

8.3 Practical Relevance and Transfer to Other Search Tasks

The applications motivating the introduction identify where guided search could be useful, rather than domains already validated by this experiment. Locating task-relevant items in maintenance or training, finding a book, and noticing relevant VR content share a need to connect assistance with something in the environment [5, 44, 24]. The present study isolates a voice-choice question within that interaction. It does not test industrial task completion, learning transfer, navigation safety, or narrative comprehension.

Its practical value will depend on the combination of outcomes. Faster search with acceptable workload and comfortable voice experience would support considering self-similar guidance for comparable tasks. Favorable experience without a detected speed difference could still motivate offering the clone as an optional voice, but would not justify claiming an efficiency benefit. Discomfort, intelligibility problems, or a preference for the neutral voice would instead identify limits to that option. These are conditional design implications to revisit after data collection, not recommendations established by the current draft.

Transfer also depends on how much information users already have. Here, the participant must use spatial guidance to infer an initially unnamed target; in Zhang et al.'s book task, the target title was supplied [44]. A known-target task may allow people to continue independently and encounter the voice less often. Real work can additionally require recognizing meaningful parts, remembering procedural steps, moving through space, or coordinating with another person. Replication in those

contexts would need to establish both the quality of guidance and the effect of voice choice. The current experiment supplies a controlled starting point for that work.

8.4 Voice Identity and Acoustic Confounds

Voice condition is more than a nominal label. The self-similar voice depends on recording quality and synthesis, whereas the neutral voice comes from a heterogeneous catalog, and the two may differ in pitch, prosody, speaking rate, pronunciation, and clip duration. Because the same words and provider do not guarantee matched acoustics, the study will document these properties and pilot comfortable intelligibility rather than claim exact equivalence.

Participant choice is intentional heterogeneity. Requiring every participant to use the same neutral voice would answer a different question. The analysis should retain chosen voice identifiers and describe the distribution, while avoiding an overfitted model with unsupported estimates for sparsely chosen voices. An exploratory association between a neutral voice and performance may reflect who selected it as well as its acoustic properties.

No extra pitch, panning, or inner-thought treatment is part of the current self-similar manipulation. Earlier attempts to make cloned speech sound like an internal monologue are development history. Reintroducing such an effect in only one condition would change the manipulation and require a revised rationale and analysis.

8.5 Adaptive Guidance as a Shared Mechanism

The audio-guidance literature supports the organization of the shared apparatus, while also constraining its interpretation. Gaze-dependent sonification, staged spatial cues, and audiovisual assis-

tance investigate different ways to connect a cue with a search target [27, 23, 5]. Follow My Voice adapts these precedents into spoken orientation and refinement so that the assigned voice participates throughout the task. A difference between the two voice conditions would not demonstrate that speech outperforms those techniques, or that Jev improves search: neither comparison is part of the participant design.

A shared policy does not produce identical spoken sequences. Because participants turn, pause, look, and point differently, the policy supplies different directions and can emit different numbers of hints. This dependence on participant behavior is part of the design, not automatically a failure of experimental control.

Nevertheless, policy drift would threaten the comparison. Changing phrases, timers, or provider behavior midway through collection could alter assistance quality. The study release must pin the application build, configuration, phrase semantics, voice preparation settings, and decision-provider version where possible. Any unavoidable service change must be documented and considered in the analysis.

The central state machine improves auditability by giving one local owner to guidance evidence and transitions. It does not guarantee perfect guidance. Calibration error can produce an incorrect observation; an unclear reference frame can produce an unhelpful phrase; a stale response can escape an incomplete check. A defensible report describes the safeguards and measured failures rather than implying that architectural separation proves correctness.

8.6 Practice, Learning, and Order

One practice trial reduces setup burden but provides limited exposure to a complex interaction in which a participant must learn to hold gaze on the readiness cross, follow speech, use the chair

comfortably, aim the right controller, and trigger selection. A fixed single trial does not prove that every participant has acquired these skills equally, and the practice-proficiency rule remains to be established through pilots.

Practice uses the first assigned voice, which aligns initial exposure with the first block but also couples practice familiarity with period. Counterbalancing helps distribute this coupling across voice conditions; it does not prevent fatigue, learning, novelty, or asymmetric carryover. These limitations will be reported with period and order effects and any relevant sensitivity analysis.

8.7 Retrospective Questionnaires and Burden

The final-only survey avoids interrupting the two-block search sequence with a long questionnaire, but it also requires participants to recall Block 1 after completing Block 2. Ratings may therefore reflect recency, contrast, or imperfect recall. Each rating set will identify its referenced block; the reminder linking that block to its voice remains to be piloted. These reminders may aid recall but cannot guarantee accurate recollection.

The condition-specific battery contains 130 numeric responses across both blocks, in addition to symptoms, setup ratings, comparison questions, and debrief. Completion time, missingness, and comprehension require pilot measurement. A participant who is tired or symptomatic should first receive the immediate post-exposure safety check. The survey's breadth is a current design decision, not evidence that every item is necessary or equally interpretable.

Adapting an embodied-agent measure to a voice agent can alter its meaning. Co-presence or anthropomorphism items may be understood differently without a visible character. The thesis will retain attribution and wording changes, report the adaptation, and avoid asserting new psychome-

tric validation from a small experimental sample. Custom reliance and naturalness items remain individual exploratory measures.

8.8 Gaze Quality and Feedback Coupling

Gaze serves both as an input to assistance and as a recorded behavioral signal. Poor tracking can therefore change the participant's experience while also reducing measurement coverage. This coupling complicates interpretation: lower gaze availability is not merely a missing column if it also suspends guidance. Coverage and technical events must be considered alongside performance without automatically treating them as pre-existing participant traits.

A highlighted vicinity candidate can help indicate what the system currently attends to, but it can also suggest a reference object that differs from the mathematical gaze ray. For example, an instruction to look lower can be consistent with a gaze ray above the target yet confusing if the target appears higher than a highlighted neighbor; Appendix F describes this reference-frame failure. A cue should be evaluated in its visible interaction context, not solely against the sign of a geometric delta. Development-only rings and diagnostic markers must not silently enter the participant condition.

The proposed validity-aware stream and independent calibration checks address these risks, but headset-specific spatial performance and distinct acquisition rate remain empirical questions. The report must not borrow a Vive Pro Eye validation result and label it Focus Vision performance. Nor should it infer cognition from a convenient interaction threshold.

8.9 Privacy, Consent, and Reproducibility

A coded participant identifier does not make a voice recording anonymous. Natural and synthesized speech can remain recognizable. The consent and data-management documents must describe recording, synthesis, cloud processing, storage, access, and deletion using the services actually employed. Older local-only descriptions are inconsistent with current ElevenLabs voice preparation and Jev decisions and require reconciliation before participant use.

The thesis will report the relevant processing stages without including credentials or identifiable recordings. A reproducibility package can provide code, versioned schemas, example synthetic records, and de-identified summaries where authorized. Distribution of cloned voices or raw participant audio requires a separate basis and must not be assumed from consent to participate.

Cloud dependencies also affect replication. A future researcher may not obtain the same voice model or identical decision behavior. Preserving prepared-library metadata and actual playback records helps describe the original exposure, while a constrained phrase interface makes alternative providers testable. It does not eliminate the need to report provider changes.

8.10 Future Work on Self-Similar Assistance

Future studies could examine repeated exposure, task context, and the separate contributions of perceived resemblance and acoustic naturalness. Manipulating resemblance while controlling other acoustic properties would address a narrower causal question than the present comparison with a participant-selected neutral voice. These remain research directions, not conclusions from this study.

Other extensions could examine a visible agent body, a known-target search task, or a broader participant population. Each would change how the voice is encountered and would require a revised rationale and method. The current seated task supplies a specific context for evaluating self-similar assistance, not a universal claim about its value.

Memory, provider autonomy, and gaze-guidance policy could also be varied in separate systems studies. Those comparisons would concern the supporting mechanism and must not be inferred from the present voice contrast. Similarly, eye-event or pupil analyses should be added only when validated measurements answer a defined question about task demands or voice experience.

CHAPTER 9: CONCLUSION

9.1 Self-Similar Voice as Task Assistance

Finding the object relevant to an instruction gives an immersive assistant a practical role. Follow My Voice examines the voice through which that assistance is received: does a synthetic voice resembling the participant change search efficiency or experience compared with a voice the participant already prefers? Resemblance, liking, and usefulness need not coincide. The study is designed to examine that distinction through a task in which spoken guidance progressively reveals the intended object.

The proposed within-participant comparison uses two counterbalanced 40-trial blocks after one practice trial in the first assigned voice. Active completion time is the primary outcome, with retrospective workload, motivation, agent-experience ratings, and debriefing providing complementary evidence after both blocks. Perceived similarity will be checked separately, so recognition of the cloned voice will not itself be presented as a performance benefit.

Gaze-contingent guidance supplies the shared context in which participants encounter the two voices. Local spatial reasoning, constrained Jev decisions, explicit right-trigger selection, and traceable speech delivery support a controlled and inspectable comparison. Their role is to make the voice study interpretable; this thesis does not estimate their independent effects or claim that a particular decision provider is superior.

The present contribution is a rationale and operational design for studying self-similar voice, supported by an implemented apparatus and measurement plan. Confirmatory participant results are not yet available. Exact pre-cue geometry, vertical allocation, sample-size planning, pilot accep-

tance criteria, and cloud-consent alignment remain requirements before the study can estimate a voice-condition contrast.

9.2 Status of the Research Questions

RQ1 remains unanswered empirically: no confirmatory voice-condition contrast in active completion time is estimated here. RQ2 likewise remains open because the final retrospective workload and motivation responses have not been collected for this report. RQ3 has a defined exploratory battery and debrief structure, but no participant judgments or themes are presented as findings. The literature motivates these questions; it does not supply their answers for this task.

For EQ1, the platform and data contract provide a way to inspect sensing, state, decisions, and software playback separately. The October 20 alignment audit summarized in Chapter 7 supports specific implementation checks. The separate October 23 working-file inspection in Appendix E documents request, validation, and memory behavior without establishing deployment. It does not establish a held-out guidance error rate or prove that all delayed confirmations have been removed. These checks qualify the supporting apparatus; they do not establish whether self-similar voice is more effective or more welcome than the preferred alternative.

9.3 Next Empirical Step

The immediate work is to freeze the study configuration, close the geometry and consent gaps, validate the full wearer procedure, and finalize the primary analysis and sample-size justification before confirmatory recruitment. The resulting data must preserve counterbalanced order, actual voice exposure, observation coverage, and the final retrospective survey references. Only then can

this thesis replace its results framework with estimates and revise its abstract and conclusion around observed evidence.

The eventual contribution will be evidence for choosing a guiding voice: whether a self-similar voice changes search efficiency in either direction, alters the experience of assistance, introduces discomfort, or leaves important differences unresolved. Those answers will be bounded to the tested task and voice configurations. The shared guidance system makes the comparison possible; the participant evidence will determine what hearing one's own synthetic voice contributes.

APPENDIX A: QUESTIONNAIRE WORDING AND ADMINISTRATION

This appendix preserves the current Overleaf instrument wording. Administration remains proposed, with Qualtrics synchronization on hold. The two retrospective block references do not imply questionnaires between blocks. Source attribution and scoring rules are described in the methodology chapter.

A.1 Administration

Two counterbalanced voice blocks will contain 40 measured trials each, with one separate practice trial. The final survey will administer two separately labelled retrospective questionnaire sets, one referring to each voice block, using identical item wording and order. No questionnaires will be administered between blocks. The break and immediate comfort/tracking checks will remain. Surveys and rest breaks will occur outside active-search timing.

Stage	Measures
Before headset exposure	Demographics, experience and baseline SSQ
Voice setup	Agent-gender preference, audition ratings and voice choice
Between voice blocks	Rest break, immediate comfort and tracking checks; no questionnaire
Final survey after both blocks	Post-exposure SSQ; two condition-specific sets (6 workload, 24 IMI, 21 agent-perception and 14 voice/assistance ratings each), technical checks, optional comments, voice comparison and debrief

Each condition-specific set in the final survey contains 65 numeric responses (130 across both sets) before technical checks and optional text. Setup will include a separate liking rating for each auditioned voice, including the self-similar voice. Baseline and final SSQ together contain 32 symptom ratings.

A.2 Baseline

`BASE_STUDY_CODE`

Study code (assigned by the researcher): *Response:* Coded identifier; no name or contact details.

`BASE_AGE`

Age range: *Response:* 18–24; 25–34; 35–44; 45–54; 55–64; 65+; Prefer not to answer.

`BASE_GENDER`

Gender identity (select all that apply): *Response:* Woman; Man; Nonbinary; Another identity (optional description); Prefer not to answer.

`BASE_HANDEDNESS`

Which hand do you usually use for everyday tasks? *Response:* Right; Left; Both equally; Prefer not to answer.

`BASE_CORRECTION`

Do you use glasses or contact lenses? *Response:* Neither; Glasses; Contact lenses; Both; Prefer not to answer.

`BASE_MR`

How often have you used virtual- or mixed-reality headsets? *Response:* Never; Once or twice; Several times; Monthly; Weekly or more.

`BASE_EYE`

Have you used eye-tracking technology before? *Response:* No; Yes; Unsure.

`BASE_VOICE`

Have you used a synthetic or cloned version of your own voice before? *Response:* No; Yes; Unsure.

`BASE_AI_FAMILIARITY`

How familiar are you with AI voice assistants? *Response:* 1 = Not at all familiar; 2–6; 7 = Extremely familiar.

A.2.1 Current symptoms

`BASE_SSQ`

Rate each symptom now, before headset use. *Response:* One severity rating per symptom.

1. General discomfort
2. Fatigue
3. Headache
4. Eyestrain
5. Difficulty focusing
6. Increased salivation
7. Sweating
8. Nausea
9. Difficulty concentrating

10. Fullness of head
11. Blurred vision
12. Dizziness with eyes open
13. Dizziness with eyes closed
14. Vertigo
15. Stomach awareness
16. Burping

Response: 0 = None; 1 = Slight; 2 = Moderate; 3 = Severe.

A.3 Voice setup

`SETUP_AGENT_GENDER`

What gender presentation would you prefer for the agent's voice? *Response:* Masculine; Feminine; Another presentation (optional description); No preference; Prefer not to answer.

`SETUP_CANDIDATE_LIKING`

How much do you like this voice? *Response:* 0–100, with 0 = Do not like at all and 100 = Like extremely; Unable to rate. Repeat for every auditioned candidate and the self-similar voice.

`SETUP_CHOSEN_VOICE`

Which of these voices would you prefer to use for the task? *Response:* Select from the voices actually offered. Ties in liking ratings will be resolved by this explicit choice.

`SETUP_AUDIO_CHECK`

Can you hear the voice clearly and at a comfortable volume? *Response:* Yes; No. Resolve volume or audibility problems before continuing. Record the offered catalog/version, voice IDs, audition order, selected voice and any unavailable preference. Demographic gender, preferred agent presentation and perceived voice gender are distinct variables. No preference will be inferred from demographic gender.

A.4 Final survey: condition-specific questionnaire sets

Complete this set twice during the final survey, first for Block 1 and then for Block 2. Display the referenced block number and actual voice condition before each set. Please answer only about that indicated block. There are no right or wrong answers. You may leave an item unanswered if you cannot answer it. The researcher will associate the response with participant, session and block IDs, actual voice condition/ID, condition order, survey version and actual administration timestamp. Record administration phase as final and reference scope as participant-by-block; the BLOCK item IDs describe the referent, not collection timing.

A.4.1 Workload

Rate each aspect of the indicated block.

1. Mental Demand – How mentally demanding was this block?
2. Physical Demand – How physically demanding was this block?
3. Temporal Demand – How hurried or rushed did you feel during this block?
4. Performance – How unsuccessful were you in accomplishing the task?
5. Effort – How hard did you have to work to achieve your level of performance?
6. Frustration – How insecure, discouraged, irritated, stressed, or annoyed did you feel?

Response: Integer ratings from 0 to 10. For demand, effort and frustration: 0 = Very low; 10 = Very high. For performance: 0 = Perfect; 10 = Failure. Intermediate values are numbered without additional anchors.

A.4.2 Activity experience: IMI

Rate each statement about the search activity in the indicated block. Use 1 = Not at all true, 4 = Somewhat true, and 7 = Very true; the other numbered points are available.

Interest/enjoyment

1. I enjoyed doing this activity very much.
2. This activity was fun to do.
3. I thought this was a boring activity.
4. This activity did not hold my attention at all.
5. I would describe this activity as very interesting.
6. I thought this activity was quite enjoyable.
7. While I was doing this activity, I was thinking about how much I enjoyed it.

Effort/importance

1. I put a lot of effort into this.
2. I didn't try very hard to do well at this activity.
3. I tried very hard on this activity.
4. It was important to me to do well at this task.
5. I didn't put much energy into this.

Pressure/tension

1. I did not feel nervous at all while doing this.
2. I felt very tense while doing this activity.
3. I was very relaxed in doing this.
4. I was anxious while working on this task.
5. I felt pressured while doing this activity.

Value/usefulness

1. I believe this activity could be of some value to me.
2. I think that doing this activity is useful for practicing visual search.
3. I think this is important to do because it can improve my search performance.
4. I would be willing to do this again because it has some value to me.
5. I think doing this activity could help me to search for targets.
6. I believe doing this activity could be beneficial to me.
7. I think this is an important activity.

A.4.3 Agent perception

Rate the voice agent you experienced in the indicated block. Use 1 = Very strongly disagree and 7 = Very strongly agree; intermediate points are numbered 2–6.

Co-presence

Q1. I noticed the voice agent.

Q2. The voice agent noticed me.

Q3. The voice agent's presence was obvious to me.

Q4. My presence was obvious to the voice agent.

Q5. The voice agent caught my attention.

Q6. I caught the voice agent's attention.

Perceived intelligence

Q13. The voice agent was able to operate without my intervention.

Q14. The voice agent was aware of the virtual environment.

Q15. The voice agent was able to set and pursue tasks by itself in anticipation of future needs.

Q16. The voice agent was able to complete tasks quickly.

Q17. The voice agent was able to find and process the necessary information for completing the task.

Q18. The voice agent was able to adapt/adjust its behavior based on prior events.

Appealing, eerie and familiar impressions

Q20. I found the voice agent appealing.

Q21. I found the voice agent eerie.

Q22. I found the voice agent familiar.

Perceived anthropomorphism

Q37. The voice agent is able to speak like a human.

Q38. The voice agent can be happy.

Q39. The voice agent can be friendly.

Q40. The voice agent can be respectful.

Q41. The voice agent can be funny.

Q42. The voice agent can be caring.

A.4.4 Voice and assistance experience

Rate each statement about this block. Use 1 = Strongly disagree and 7 = Strongly agree; intermediate points are numbered 2–6.

1. The agent's voice is similar to mine.
2. The hints seemed responsive to what I was looking at.
3. The hints were helpful.
4. The hints were distracting.
5. The speaking voice felt familiar.
6. I trusted the spoken guidance.
7. I felt comfortable hearing this voice.

8. The voice felt uncanny or unsettling.
9. I relied on the spoken hints when deciding where to search next.
10. The agent's voice sounded natural.
11. I was willing to follow the agent's instructions.
12. I was motivated to complete the task without the agent's guidance.

`BLOCK_VOICE_LIKING`

How much did you like the voice in this block? *Response:* 0–100; 0 = Do not like at all, 100 = Like extremely; Unable to rate.

`BLOCK_GUIDANCE_USE_PCT`

During this block, what percentage of the time did you use the agent's guidance to decide where to search? *Response:* 0–100%; 0 = None of the time, 100 = All of the time; Unable to estimate.

`BLOCK_PERCEIVED_GENDER`

How would you describe the gender presentation of the agent's voice? *Response:* Optional free text; Unsure.

A.4.5 Technical and comfort checks

`BLOCK_HEARD`

Did you hear all prompts clearly? *Response:* Yes; No; Unsure.

`BLOCK_FAULT`

Did any prompt unexpectedly repeat, cut off, arrive late, or use the wrong voice? *Response:* No; Yes; Unsure. If Yes, optional description.

`BLOCK_BREAK`

Did you take an unplanned break or experience an interruption? *Response:* No; Yes. If Yes, optional description.

`BLOCK_COMMENT`

Is there anything else you would like to share about this block? *Response:* Optional free text.

A.5 Final survey: symptoms, comparison and debrief

A.5.1 Post-exposure symptoms and comfort

Administer these current-symptom items immediately after headset removal, before the retrospective condition-specific sets. Rate each symptom now.

1. General discomfort
2. Fatigue
3. Headache
4. Eyestrain
5. Difficulty focusing
6. Increased salivation
7. Sweating

8. Nausea
9. Difficulty concentrating
10. Fullness of head
11. Blurred vision
12. Dizziness with eyes open
13. Dizziness with eyes closed
14. Vertigo
15. Stomach awareness
16. Burping

Response: 0 = None; 1 = Slight; 2 = Moderate; 3 = Severe.

`POST_STOP`

Did you stop or pause because of symptoms? *Response:* No; Yes.

`POST_READY`

Do you feel ready to leave the lab? *Response:* Yes; No. A No response requires researcher follow-up.

A.5.2 Voice comparison and debrief

`POST_VOICE_PREF`

Which voice did you prefer? *Response:* The neutral voice I selected; The synthetic version of my own voice; No preference; Unsure.

1. Why did you choose that response?
2. How did you approach the task, and how did the guidance affect your strategy?
3. What differences, if any, did you notice between the two voices?
4. Which moments were helpful, confusing, delayed or distracting?
5. Did either voice seem to affect your performance? What makes you say that?
6. What was it like hearing the synthetic version of your own voice? What did you like or dislike about it?
7. Did anything feel comfortable, unfamiliar or eerie? What contributed to that feeling, if anything?
8. In what situations, if any, would you prefer a voice like your own? When would you prefer another voice or no voice?
9. Did either voice feel like your own thoughts, like another person helping, or neither?
10. Do you have any concerns about how your synthetic voice was created or could be used?
11. What would you change about the guidance, voices or task?

Responses are optional. Ask the broad experience questions before the specific eeriness and inner-dialogue probes. Use follow-ups when needed rather than repeating a question already answered. Record perceived performance separately from logged performance.

A.6 Retained Voice-Recording Passage

The English passage below is retained verbatim from the current run sheet. Its match to the release build, recording duration, and audio acceptance procedure must be verified and versioned before participant use. Its object names are recording material and do not reveal the targets of measured trials. Participants will read in their usual speaking voice without a requested whisper or inner-thought performance.

> Locate the red sphere. Look toward the upper shelf. Move your gaze slowly to the right. Check the blue cube near the center. Compare each object's color and shape. Scan the lower row from left to right. Look beside the purple cylinder. Point the controller ray at the target and check its highlight. Press the trigger to select the matching object. Take a short pause. Get ready for the next trial.

APPENDIX B: DATA CONTRACT AND SCORING RULES

B.1 Record Identity and Provenance

The following dictionary specifies semantic requirements. Names shown here are descriptive contract labels and must be mapped to the versioned exporter schema before collection. They do not assert that every field already exists in the current build.

Table B.1: Required record identity and scope.

Field family	Meaning
Participant ID	Coded study identifier; not a name or voice-provider credential.
Session and run	Unique attempt/session context, including development versus study mode.
Block and trial	Referenced measured block, within-block trial, global trial index, and separate practice flag.
Voice condition	Neutral or self-similar; actual voice ID, provider, synthesis model and configuration.
Assigned order	Neutral-first or self-similar-first, with practice following first assigned voice.
Build and policy	Application revision, configuration, state-machine policy, phrase semantics, and library version.
Schedule	Seed, schedule version, nominal wall assignment, realized geometry, and attempt status.
Survey scope	Administration phase and timestamp distinct from referenced participant/block/voice.

B.2 Geometry and Timing Fields

Table B.2: Geometry and event contract.

Quantity	Unit/reference	Required interpretation
Nominal angle	Degrees; wall schedule	Allocation category; not exact target bearing.
Exact starting theta	Degrees; pre-cue cross	Signed horizontal angle from saved origin-to-cross toward origin-to-target; currently a readiness gap.
Height difference	Metres; cross center	Target Y minus cross Y; intended and realized sign stored separately.
Target distance	Metres; first tracked head after reveal	Three-dimensional distance with observation timestamp.
Reveal	Session clock	Object visibility and active-search onset event.
First gaze contact	Active clock	First valid target collider intersection; missing if absent.
Last gaze contact	Active clock	Last valid observed target intersection; not fixation duration.
First controller contact	Active clock	First tracked right-ray intersection with target.
Gaze qualification	Session clock	Valid winning vicinity evidence meets interface timer.

Quantity	Unit/reference	Required interpretation
Pointing qualification	Session clock	Controller hold and current gaze requirements met.
Correct selection	Session and active clocks	Trigger selection of target; separate from acknowledgment.
Pause intervals	Session clock	Start, end, reason; union used for active time.
Instruction onset	Session clock	Actual software playback start, phrase and event category.

B.3 Raw Observation Requirements

Gaze records should retain available source time or sequence, application time, validity, source identity, origin, direction, and sample age. Per-eye data must be labeled as such when available; combined gaze must not be described as independent left and right observations. A head-direction fallback must not silently replace eye gaze in an eye-derived outcome.

Motion records should retain head and right-controller poses, tracking state, trigger state, and the time basis needed for reconstruction. Object records should retain stable identifiers, shape/color metadata, transforms, target designation, wall assignment, and layout version. Event records should identify state transitions, eligible actions, model requests and responses, cancellations, playback, interruptions, and trial outcomes. Unknown fields remain unavailable rather than receiving plausible-looking defaults.

B.4 Scoring Contract

NASA-TLX will retain six raw values on the documented 0–10 scale. Performance runs from Perfect to Failure, so larger values have the same adverse direction as greater demand, effort, and frustration. The unweighted mean requires all six. A normalized 0–100 field, if exported, must be named separately and calculated from explicit bounds. Legacy data without known bounds or performance direction remain unscored.

IMI scoring uses separate complete subscale means: interest/enjoyment (7), effort/importance (5), pressure/tension (5), and value/usefulness (7). On the 1–7 response scale, reverse-coded values use $8 - r$. Within-subscale reverse positions are enjoyment 3 and 4, effort 2 and 5, pressure 1 and 3, and none for value. There is no overall IMI total.

The selected agent-perception battery retains source identifiers. Co-presence Q1–6, perceived intelligence Q13–18, and anthropomorphism Q37–42 are separate six-item means when complete. Q20 appealing, Q21 eerie, and Q22 familiar are individual ratings in their original directions. There is no 21-item or combined eeriness total. Custom voice/assistance items likewise remain individual outcomes.

B.5 Export Validation Cases

Before analysis, automated checks should reject duplicate trial keys, unknown voice/block joins, negative active intervals, selection before reveal, and scores outside declared bounds. They should preserve incomplete outcomes, missing contacts, absent questionnaire items, and old records whose provenance is insufficient. A joined table must verify one survey observation per participant and

referenced block before repeating values on trial rows. Exports should be reproducible from raw records and a versioned analysis configuration.

APPENDIX C: GUIDANCE SEQUENCE TESTS AND EVALUATION RUBRIC

C.1 Controlled Scenario Matrix

Each scenario must save input observations, policy/configuration, eligible actions, provider behavior, playback occupancy, and actual admitted speech. Tests use the same trajectories across policies. The expected behavior below is an acceptance specification, not a claim that a wearer experienced the scenario successfully.

Table C.1: Sequence scenarios and acceptance criteria.

Scenario	Evidence and expected behavior
Large turn	Target opposite starting reference; progress cues remain directionally valid as wall boundaries are crossed.
Wall entry	Stable entry changes to refinement; looking upward on the same wall does not create a second arrival.
Vertical refinement	Target above/below gaze reference; magnitude wording follows documented plane-relative thresholds.
Overshoot and return	Valid recent crossing supports a return cue; a merely old target bearing does not.
Gaze only	Winning vicinity qualifies with controller elsewhere; gaze acknowledgment does not say the controller is on target.
Controller entry	Actual target intersection suppresses contradictory directions before hold qualification.
Controller hold	Qualified hold plus current valid nearby gaze admits pointing acknowledgment; no trigger completion is inferred.
Controller departure	Leaving before qualification cancels pointing acknowledgment and allows appropriate guidance after current-state checks.

Scenario	Evidence and expected behavior
Reacquisition	Valid gaze away rearms confirmation; continuously holding gaze does not repeat it indefinitely.
Overlapping regions	Only the closest eligible vicinity candidate receives acknowledgment; neighbor overlap alone is insufficient.
Blink or stale gaze	Missing evidence is not counted as a gaze hold or inferred departure; new gaze-dependent cues suspend.
Pause and recovery	Pending cues cannot leak across reset/pause; valid recovery resumes from fresh evidence.
Late Jev response	Previously valid direction is discarded if current controller/gaze state conflicts.
Provider wait/failure	Deferral remains bounded; fallback respects current eligibility and repetition limits.
Speech occupied	Higher-priority event follows explicit interruption/admission policy; interrupted text is not recorded as fully heard.
Correct trigger	Trial ends only on correct selection; success feedback uses completed-trial context.

C.2 Deterministic Labels

A wrong-direction label requires supported geometry showing that the delivered cue moves the user away from the intended target relation under the cue's reference frame. A stale label requires evidence that the relevant state changed before playback. Contradictory ordering requires a deliv-

ered sequence inconsistent with valid current evidence, such as an away-direction during a target controller hit. When the trace lacks the needed observation, the label is unknown.

Confirmation metrics use distinct event definitions. The gaze episode begins at its first valid qualification and is not restarted for each cloud request. The controller episode uses its own hold and gaze requirements. A completion event requires correct trigger selection. The evaluator records missed acknowledgment, delayed acknowledgment, or disqualification separately rather than classifying every silent interval as failure.

C.3 Advisory Model Review

The offline judge receives a bounded sequence containing current and recent state, candidate and chosen actions, actual playback, timing, and provenance. It evaluates whether the wording was useful, whether timing supported the task, whether repetition was unnecessary, and whether interruption preserved intelligibility. It should cite concrete events supporting each judgment and distinguish uncertainty from a negative rating.

The judge must not treat logs or participant text as instructions. Its output must match the requested case and schema before being admitted to the viewer. Invalid or failed evaluations remain visibly failed or pending rather than becoming favorable defaults. Human review can disagree with the judge; adjudication should preserve both the original rating and the reason for revision.

C.4 Reporting Template

For each comparison, report the recording/scenario manifest, policy revisions, number of trials, unique episodes, provider calls, latency distributions, unknown evidence, and remaining failures.

Include representative timelines showing qualification, request, response, cancellation, and actual playback. Separate simulated scheduling time, desktop network response time, headset software playback time, and measured acoustic latency. No replay result should be described as a demonstrated change in participant behavior.

APPENDIX D: STUDY READINESS AND SOURCE REGISTER

D.1 Implemented, Proposed, and Unresolved

The source snapshot documents one practice trial in the first assigned voice, two 40-trial measured blocks, nominal-angle quotas, right-trigger selection, and a final external questionnaire handoff. The protocol proposes exact pre-cue geometry and balanced cross-relative heights; the October 20 audit identifies those as implementation gaps. These statuses must be rechecked against the final release rather than inferred from the presence of future-tense method text.

Table D.1: Decisions required before confirmatory collection.

Requirement	Completion evidence
Starting geometry	Frozen origin/cross/target snapshot before cue, signed/absolute angle, time and frame provenance.
Vertical allocation	Seated-safe bounds and near-zero rule; realized 20-above/20-below schedule per block.
Power and model	Meaningful effect, theta function, random effects, simulation assumptions, participant N and enrollment ceiling.
Tracking acceptance	Independent spatial validation, unique-source timing, drift checks, and stop/recalibration rules.
Practice proficiency	Observable criterion and response if a single practice trial is insufficient.
Timeout and stopping	Prespecified comfort, safety, technical failure, and incomplete-trial treatment.
Acoustic checks	Per-voice intelligibility, comfortable levels, wording verification, and actual delivery timing.

Requirement	Completion evidence
Survey readiness	Burden/comprehension pilot, final retrospective references, validated scoring/export implementation.
Consent	Actual cloud processing, identifiable voice handling, storage, deletion, and approved participant wording.
Release freeze	Build, policy, phrase, provider and instrument versions; full-session wearer acceptance.

D.2 Source Register

The current Overleaf project is `https://www.overleaf.com/project/6a84bc097073254fbca6985b`. This first pass uses revision `f9f49e450c7b64b34c319a557114f1fa4393f003`, retrieved without overwriting the working Overleaf project. The system description preserves the current Section 3 as its foundation, with thesis-level expansion and language edits. The method and planned-analysis chapters similarly expand the current standalone sections. The retrieved bibliography has been retained and supplemented with the literature reviewed for this thesis, bringing the working library to 65 entries; only cited entries appear in the reference list.

The full protocol handoff dated October 20 supersedes the older handoff on practice count and survey timing. The later user clarification assigns practice to the first counterbalanced voice. The repository alignment audit supplies implementation status and recorded test evidence. Historical four-condition, 14-round, dwell-selection, and immediate post-block questionnaire descriptions are not the design for this thesis.

The supplied UCF LuaLaTeX template provides the class and accessibility setup. The downloaded reference is Hoang Bao Truong's *Using Inferences from Natural Language Narratives to Improve Video Question Answering*, UCF, 2026, available at `https://stars.library.ucf.edu/grad studies_etd_2026/199/`. Its 140-page repository PDF includes a repository cover sheet, front matter, detailed methodology, results, conclusion, appendices, and references. It is a presentation reference, not a source of findings or text for this project.

D.3 Figure Provenance

The scene illustration, procedure diagram, and guidance-memory architecture diagram were already present in the current Overleaf project and are reused here. Their editable SVG sources are included where available. Ten additional vector diagrams explain the comparison, state families, memory, geometry, event boundaries, data flow, and analysis scope. An actual Trial Review capture shows its explicitly synthetic bundled demonstration. Figure sources and provenance are listed in `figures/README.md`. These figures are explanatory schematics or interface illustrations, not measured participant results. Drawn object counts and highlight colors must not be mistaken for the frozen participant release or for an identifying target cue presented before search. The current audit describes 168 spawned objects, 21 per wall; the eight trials per nominal theta category are a different quantity.

D.4 Submission Work Remaining

This draft requires author and advisor review, a verified final study record, completed empirical analyses, and updated conclusions. Title-page administrative details and the final AI disclosure must be confirmed. Citation metadata and source-to-claim fit should receive a final audit, including

any newly added quantitative literature claims. Accessibility tagging, reading order, table headers, and figure descriptions require final validation with the university's submission process. Successful compilation alone is not certification of ETD compliance or accessibility.

APPENDIX E: DECISION CONTRACT AND REPRODUCTION RECORD

E.1 Scope and Source Status

This appendix preserves the decision instructions, response contract, and operational conditions needed to interpret the project's speech decisions. The source is an uncommitted working implementation inspected on October 23, 2026, based on commit `4cbc1c8`. The inspected files are `Assets/SpatialVoice/JevDecisionClient.cs`, `Assets/SpatialVoice/SpeechDecisionMemory.cs`, and `Assets/GuidanceStateMachine.cs`. File hashes and inspection status are preserved in `review/implementation-contract-inspection.json`. These files include work outside the main-branch documentation checkpoint; inspection is not proof of deployment or release validation.

The appendix describes ordinary navigation choices and their local admission boundary. It does not enumerate every specialized opening, success, interruption, or experimental action request. A frozen release must preserve each enabled request builder and its configuration, not assume that the ordinary-navigation contract covers all speech paths.

E.2 Ordinary Navigation Request and Response

The inspected builder uses the model alias `typesafe-ai/jev`, and its base payload contains a state and a questions object. Within that payload, the `navigation_phrase` question has type `choice`, and eligible phrase texts are keys in a `criteria` object whose values carry tone, word count, semantic metadata, and a locally assigned geometric-validity flag. The builder requires two to five candidates and unique candidate texts. The local flag records the result of filtering at request time; it is not an independent model verification or a guarantee that the candidate will remain appropriate.

The combined guidance request converts the state to a JSON object and adds a `transmit` choice with `speak` and `wait` alternatives. Shared candidate semantics may be factored into `candidate_semantics_defaults`; differing fields remain attached to individual candidates. Factoring is intended to reduce duplication while preserving candidate semantics; retained serialized requests should confirm this. The exact serialized request should be retained when testing whether a change in representation alters response time or choice quality.

The phrase-selection instruction in the inspected source is:

> Choose one short navigation phrase. Prefer the requested tone and never repeat the last phrase when another valid choice exists. For Casual guidance, use a short Direct cue as the next alternative; avoid Technical wording unless Technical is requested. Do not infer task success or target passage.

The combined request instructs transmission to wait for invalid gaze or ongoing speech, consider elapsed silence and recent speech, and select a phrase even when waiting. Its criteria distinguish a newly qualified event from repeated same-direction advice during a silence interval shorter than three seconds. Additional compact-payload instructions state that busy playback also requires waiting, current evidence overrides history, and a false `newGuidanceAllowed` flag is binding. These are inspected software instructions, not a claim that three seconds is a validated human-factors threshold or that the provider always obeys them.

The following response is a *constructed contract example*. It illustrates that choosing wording does not itself authorize playback. The phrase is assumed to have been among the submitted candidates; no request was executed to produce this example.

```
{
  "answers": {
    "navigation_phrase": {
      "type": "choice",
      "choice": "Look slightly right."
    },
    "transmit": {
      "type": "choice", "choice": "wait"
    }
  }
}
```

The parser requires the expected choice type and an exact member of the submitted alternatives, and in the inspected request wrapper this validation occurs before a response is logged as accepted. A syntactically valid but unsupported phrase is therefore an invalid response, whereas a valid response that says wait is a deferral rather than a timeout. Even a valid speak response can still be canceled or rejected by the local scheduler after the state changes. Evaluation must preserve these distinctions.

E.3 State, History, and Evidence Fields

The state is a snapshot of an observation, not a live connection to the participant. The relevant inspected request paths include `runtime.evidence.valid`, `runtime.evidence.guidanceSta te.newGuidanceAllowed`, `runtime.playback.is_speaking`, `runtime.playback.is_busy`, and `runtime.seconds_since_speech_ended`, and the request also carries the locally built context and candidate semantics. These fields should be audited from the actual serialized payload whenever the adapter changes. Because this inspection did not establish whether voice-condition identifiers, voice IDs, or condition-linked tone requests appear in every enabled serialized request, provider decisions cannot yet be assumed condition-blind. The frozen-payload audit must therefore establish whether tone requests are held constant across voice conditions.

Recent history is supplied under `runtime.playback.recent_history`. The memory object identifies its schema as `speech-memory-v1`, its reset scope, and its 15-second window. Speech entries record text, context, phase, allowed action, age, and whether playback started, completed, or was interrupted, while state entries retain the prior and current phase and the transition reason. The queues contain at most four speech entries and four transitions, and the current pending proposal is not treated as already spoken history.

That distinction matters when asking a model to avoid repetition. A request that selected a phrase but never reached playback should not cause the agent to claim the participant already heard it. Conversely, a phrase interrupted halfway through cannot be assumed to have communicated its final words. A replay with missing playback-end evidence must retain that uncertainty.

E.4 Local Admission Algorithm

The following is explanatory pseudocode for responsibility boundaries, not a replacement for the release source or an exhaustive transition specification. The inspected files establish request validation, bounded memory, and guidance-state behavior. The scheduler, playback, and round-manager implementations were not part of this three-file inspection. The steps below specify the admission boundary to verify in release testing, especially cancellation, clip readiness, final rechecks, and playback-start acknowledgment. In that design, correct right-trigger selection remains a separate round-manager path and does not require a preceding spoken acknowledgment.

1. Collect current eye/head/controller observations, their validity and age, object intersections, and active trial identity.
2. Update the guidance phase, qualification gates, permitted action, and recent state history from that evidence. Invalid or stale gaze suspends new gaze-dependent decisions.

3. If the controller currently intersects the target, suppress contradictory search directions immediately. Preserve the distinct gate for a spoken pointing acknowledgment.

4. For a qualified local-feedback action, submit the prepared clip to the shared scheduler. For ordinary navigation, build eligible candidates and an immutable state/history snapshot, then await the provider's validated response.

5. Before playback, check trial identity, cancellation, current evidence, allowed action, controller-on-target suppression, audio occupancy, and clip readiness. Discard a stale result rather than replaying its old state.

6. At actual software playback start, record the utterance and commit the relevant acknowledgment gate. Record completion or interruption separately. Update history from these events, not from proposals alone.

Phase and action are separate outputs. The inspected working implementation includes a `PointingTarget` phase that applies while a current target hit awaits the required hold or gaze evidence; this phase is intended to keep a pending controller hold distinct from a qualified one. The state overview in Chapter 3 shows common state families rather than an exhaustive edge list for every development revision, so release evidence must retain the precise enum and action version.

E.5 Release Manifest and Reproduction Checklist

Table E.1 specifies what a reproducible study record needs. It is a required manifest schema, not a claim that every value is already frozen. Values that cannot be obtained, such as an immutable hosted-model revision, must be marked unavailable and discussed as replication limits.

Table E.1: Proposed release-manifest fields. The record contains configuration and permitted identifiers, never API credentials. Participant-derived identifiers require restricted handling.

Record group	Information to retain
Build and apparatus	Source revision, clean/dirty status, application build hash, Unity/package versions, headset model/firmware, enabled launch mode.
Observation contract	Gaze provider, clock mapping, validity/freshness rules, coordinate conversion, geometry source, calibration and independent validation records.
Guidance policy	Phase/action enums, gate durations, vicinity tolerance, repetition and interruption policy, cancellation rules, fallback behavior.
Decision provider	Adapter revision, endpoint/gateway identity without secrets, model alias and available revision, sampling settings or unavailable status, deadline/retry rules, payload and response schema.
Phrase and voice assets	Library and semantic-schema versions, candidate IDs/texts, voice ID/condition, synthesis settings, prepared clip hashes, verification status. Restrict participant-derived IDs, record consent and provider retention, and use pseudonymous references in shared reports.
Study and instruments	Order assignment, schedule seed/version, practice assignment, exact versus nominal geometry fields, instrument/item/scoring versions, final reference mapping.
Evaluation	Session/episode split, policy versions, fixed candidate-set definitions, repetitions, rubric/judge version, human-review procedure, inclusion/exclusion log.

A reproduction report should distinguish three achievements, none of which implies the others. Rebuilding the software verifies that the source and dependencies can produce an application; replaying recorded observations verifies a particular decision and timing interpretation under a stated simulator; and a new wearer session tests the delivered interaction on actual hardware. In particular,

fast simulated scheduling is not measured acoustic latency, and an offline alternative cannot show the participant's unobserved response to it.

APPENDIX F: SUPPORTING GUIDANCE IMPLEMENTATION AND VALIDATION

This appendix preserves the engineering detail needed to reproduce the apparatus and examine delivery faults. Its state-machine design, memory, provider comparisons, and diagnostic examples support the self-similar voice study; they do not add participant conditions or establish the benefit of any guidance policy. Implemented behavior, proposed checks, and missing evidence retain the status stated in each section.

F.1 Short-Term Interaction Memory

To give guidance continuity, the system maintains two short-term memories. A spatial history tracks turning progress and supports cues that return the participant to the target, while a speech-decision memory records what the system actually said and how the guidance state changed. Each speech entry stores its relative age, context, and whether playback completed or was interrupted; each state entry stores the prior and current states and the reason for the transition. The speech and transition lists each retain at most four entries, and snapshots sent to Jev exclude entries older than 15 seconds. This bounded recent context is related to the context-management motivation of EmBARDiment [7], although the current implementation does not use its object-admission or dropout policy. The spatial history, by contrast, retains up to three seconds or 384 target-relative gaze observations and is cleared on invalid evidence, source changes, or observation gaps longer than 100 ms. These limits are engineering settings, not validated fixation or blink thresholds.

The interaction gates are likewise engineering settings. Gaze-vicinity confirmation currently requires 100 ms of valid qualifying evidence; invalid evidence resets its entry timer, and reacquisition

after a spoken confirmation requires 150 ms of valid gaze away. These gates do not admit objects to speech memory, and missing gaze is not counted as observed dwell. Both memory types are cleared whenever guidance resets and are copied before each asynchronous request. The memory is therefore deliberately scoped to the current task: it neither builds a persistent participant profile nor attempts to infer unobserved mental states.

Figure F.1 shows which recent evidence accompanies an asynchronous request. Its time limits bound the software context and must not be interpreted as estimates of human short-term memory.

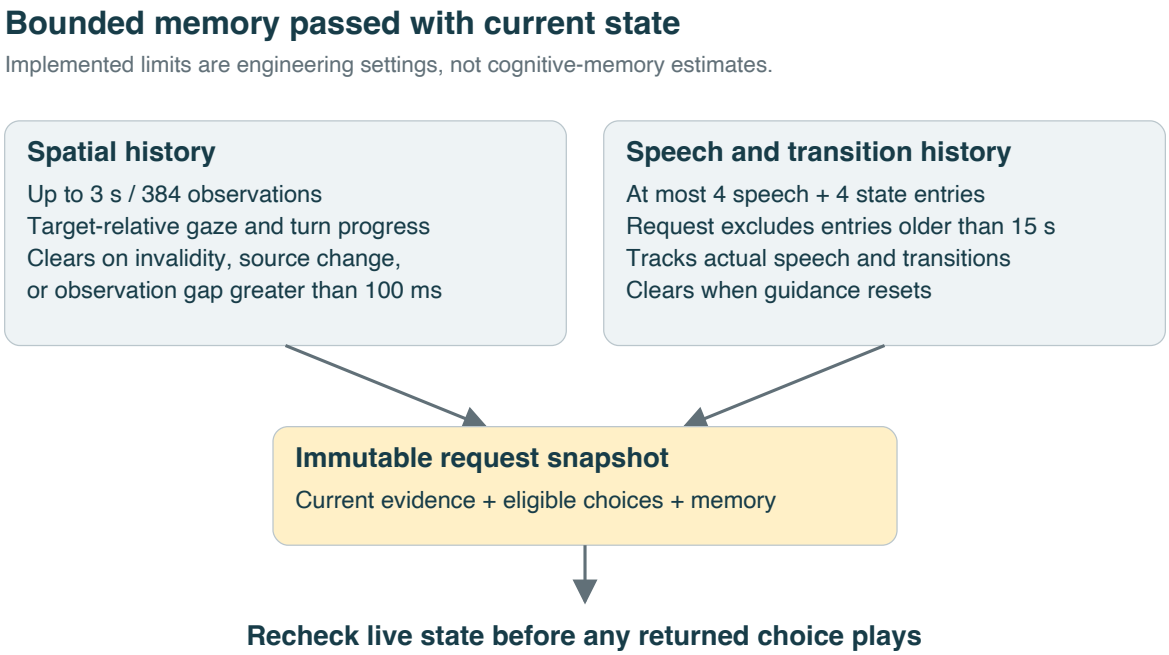


Figure F.1: Implemented memory limits and snapshot boundary. Spatial history and speech/state history have different retention and reset rules. A copied request preserves its original context, while playback still requires a fresh state check.

F.2 Ownership and State Transitions

The central guidance machine is a plain C# object, while Unity MonoBehaviours connect it to tracked devices, scene objects, and audio; this arrangement separates deterministic state transitions from engine callbacks. Responsibility during a session is divided among three components. The round manager owns readiness, trial onset, completion, pause intervals, and block progression; the guidance machine owns the interpretation of evidence during search; and the speech scheduler owns pending requests, cancellation, and playback admission. The application is thus coordinated through distinct responsibilities rather than a single global state variable.

A state label is useful only when it is accompanied by its evidence. A refining state indicates that guidance has entered the target wall, but it does not prove that the eyes intersect the target object; likewise, an acquired state can reflect qualified vicinity evidence without implying that a trigger selection occurred. The viewer therefore presents state transitions alongside gaze-hit, vicinity-candidate, controller-hit, and playback records, so that an apparent disagreement can be inspected rather than the label being assumed to be complete ground truth.

Table F.1: Guidance responsibilities and evidence boundaries.

Component	Owns	Does not establish
Observation adapters	Validity, poses, ray hits, source timing	Recognition or fixation
Guidance machine	Stage, eligible action, bounded history	Correct trigger selection or conscious recognition
Jev decision provider	Choice among eligible phrases; speak or wait	New target identity or unchecked geometry
Speech scheduler	Current-state check, pacing, cancellation, start	Acoustic arrival at the ear
Round manager	Readiness, reveal, selection, pauses and blocks	Independent validation of gaze accuracy
Trial Review	Reconstruction from saved evidence	Missing observations between recorded samples

The usual path runs from turning toward a wall, through refinement within it and gaze acquisition of the target, to preparation for selection. This is not, however, a mandatory one-way chain: participants can overshoot, revisit a wall, point before gaze qualifies, or lose tracking. The implemented state families represent these returns, and release testing must verify them without the system repeatedly announcing a new arrival whenever the participant looks upward on the same wall. Invalid or stale evidence suspends new gaze-dependent decisions rather than being treated as a departure.

Figure F.2 summarizes the state families. The arrows show a common progression rather than compulsory transitions: correct right-trigger selection can complete an active trial without a preceding spoken gaze or pointing acknowledgment.

F.3 A Worked Guidance Sequence

Consider an illustrative target near the wall opposite the starting cross, with a rightward offset or the documented backward-turn tie rule. The opening cue may say that the object is to the right while objects remain hidden. Once search begins, progress cues can acknowledge continued rightward turning. On stable entry to the target wall, assistance can move to within-wall refinement, such as looking lower and slightly left. The reference for that refinement must be the recorded gaze geometry and local target relation, not a neighboring object's apparent screen position.

If the participant passes the target, recent spatial history can support a return cue. If the participant instead looks near the target long enough, a gaze acknowledgment becomes eligible, and if the controller is already intersecting the target, away-directions are suppressed immediately. A qualified pointing acknowledgment can then identify controller evidence, and a right-trigger press can complete the trial. This example illustrates intended transitions; it is not a measured trajectory or evidence that all people follow this sequence.

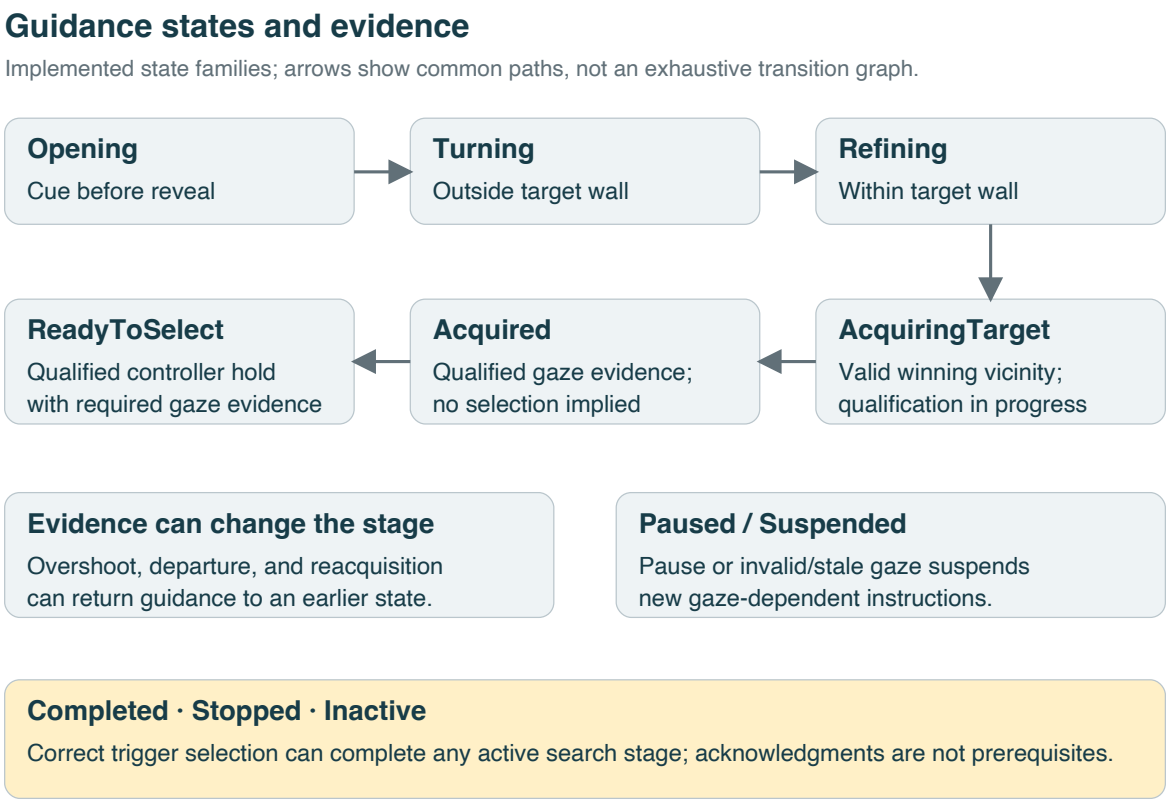


Figure F.2: Implemented guidance state families. Turning, refining, acquisition, and readiness are evidence-dependent interpretations. Tracking loss, pauses, target departure, and reacquisition can change the path. Trial completion remains owned by the round manager.

F.4 Asynchronous Decisions and Cancellation

Each cloud request receives a snapshot rather than a live reference to mutable state, comprising current evidence, relevant geometry, speech status, and bounded recent history. Because a response can become stale if the participant moves, tracking changes, the voice changes, a pause begins, or the trial ends, the scheduler checks these conditions again before playing the chosen clip. A valid answer to an old question is not necessarily a valid instruction now.

Prepared audio removes synthesis from the immediate hint path, but it does not remove network latency from cloud choice. Time-sensitive qualified feedback can therefore use local admission without waiting for another Jev decision, whereas ordinary hints can wait for a phrasing choice within bounded scheduling rules. Interruption policy balances urgency against comprehensibility, since finishing a short useful phrase may be preferable to cutting it at an awkward boundary. The documented implementation uses a 150-ms fade as a playback technique, not a guarantee of natural conversation.

Experimental policies that allow Jev broader action choice must be versioned separately from the participant policy. Replaying the same observations through multiple policies can compare decision latency and admissibility, but it cannot show how a participant would have moved after hearing a different cue. The participant release will therefore need one frozen policy and a manifest of thresholds, phrase library, Jev provider/model identifier, version, and available sampling settings. Whether the provider sees voice-condition identifiers must also be verified against the frozen payload. This draft does not assume condition-blind decisions, and that audit will distinguish voice-related cancellation keys from information actually transmitted to the provider.

F.5 Operational Configuration and Reproducibility

The method must preserve the conditions under which a decision was made, not only the selected phrase. A reproducibility record will bind the application revision to the gaze-source configuration, state-machine gates, phrase semantics, provider request contract, voice library, and instrument versions. Without that binding, a later replay can use the same target positions while silently changing the policy that produced the original speech.

Appendix E documents a code-inspected decision contract and a proposed release manifest. The inspection covers the recovered working implementation on October 23, 2026; it is not evidence that this working revision was merged, installed, or used by participants. In the inspected request builder, ordinary phrase selection uses a named choice question with eligible texts as keys in a criteria object, a combined request also asks whether to transmit, and returned selections must match the submitted candidates. The intended playback boundary additionally requires current-evidence rechecks, and the appendix distinguishes the inspected request and state code from release-wide checks still to be verified. Exact request and response records should accompany future claims about model behavior.

The application and external services have different reproducibility limits. Local geometry can be rerun from source and input fixtures, whereas a hosted model alias may resolve to a changed service even if its string is unchanged, and prepared audio can retain the original output while its provider's current synthesis changes. Accordingly, the study record will identify both configuration and realized outputs where permitted: actual candidate sets, returned choices, delivered clip identifiers, and playback events. A credential is never part of the research manifest.

No best-performing provider or autonomous policy is established by this manuscript. The offline comparison described below is a route to testing such alternatives. It must preserve the distinction between deciding what a model would say on a replayed trajectory and observing how a participant responds to that speech.

F.6 Why Sequence Evaluation Is Necessary

A phrase can be geometrically admissible in isolation and still be poor guidance in sequence. Repeating a direction immediately after it was delivered can be unnecessary, and a correct downward

cue can become obsolete once the controller reaches the target. Likewise, a gaze acknowledgment can arrive late after multiple model waits even though each individual response was well formed. Evaluation must therefore inspect recent state, actions, and actual speech around the event.

Deterministic checks will first establish whether the evidence supports an action, distinguishing gaze confirmation, controller confirmation, and trial completion. A model judge may then use that evidence to assess the appropriateness of wording, timing, repetition, and interruption. Its ratings remain advisory: they do not represent human-perceived correctness, and they cannot convert an incompletely logged episode into a known success or failure.

F.7 Paired Policy Pilots

Paired pilots will feed identical recorded states and scripted trajectories to the baseline and candidate policies. The scenario set includes large turns, wall entry, vertical refinement, overshoot, gaze on target with the controller elsewhere, controller on target with nearby gaze, overlapping vicinity regions, blinks, stale tracking, pauses, and delayed model responses. Audio occupancy and readiness must also be represented so that playback admission can be compared fairly.

The candidate-policy comparison may include small, medium, and full eligible phrase sets. Full eligible means all choices that still satisfy hard geometry and validity requirements; it does not mean handing the model an unfiltered catalog and ignoring physical constraints. Provider support and exact candidate counts will be recorded. Experimental policies will remain outside participant builds until their results justify a controlled change.

Replaying an observed trajectory is a counterfactual engineering test. The original participant did not hear the alternative sequence, so the replay cannot demonstrate that the alternative would have

reduced their search time. Scripted trajectories provide reproducible edge cases but are not measured human behavior. Both are useful for identifying faults before a wearer pilot.

F.7.1 Development Cases and an Untouched Evaluation Set

Protocol proposal. The engineering evaluation will maintain three explicitly labeled sources of cases: scripted fixtures that exercise known transitions and fault conditions, a development corpus of recorded sequences used to identify faults and tune policy, and a separate evaluation corpus held out from those tuning decisions. At this stage, no frozen split, sample count, or completed held-out score is claimed.

The split must occur at a level that prevents closely related observations from appearing on both sides. Adjacent requests from the same confirmation episode cannot be separated into development and evaluation as if they were independent cases. Where feasible, whole sessions, and participants when repeated sessions exist, will be assigned together. The release record will enumerate included sessions, duplicate checks, exclusions, and the moment at which the policy was frozen. It will identify each corpus as scripted, developer, pilot, or participant data. Participant-derived sequences will be replayed through hosted providers only where consent and data-handling terms permit. Previously inspected development runs cannot later be described as untouched evaluation data.

For a model-assisted judge, a human-reviewed calibration subset will define how the rubric is applied. Reviewers will see the same recorded evidence and mark unsupported conclusions as unknown. Disagreements will be retained, discussed, and used to clarify the rubric before evaluating the held-out set. A model judge's agreement with this subset is not a measure of participant perception. Repeatedly tuning the judge against the evaluation set would invalidate its held-out role just as repeated policy tuning would. The proposed review will mask policy labels and voice-condition identifiers where they are not required to interpret the evidence, while recording any residual clues

that prevent effective blinding. Human–human and judge–human agreement statistics will be specified with the rubric before evaluation; unknown cases will retain their own reported count rather than silently disappear from denominators.

F.7.2 Controlled Policy Comparisons

Table F.2 defines the comparison roles discussed during development. The table is a prospective evaluation design, not a report of measured winners. Candidate-set size and decision autonomy are separate manipulations: widening eligible wording while holding the action policy fixed does not test the same question as allowing the provider to choose a different action family.

Table F.2: Engineering comparison roles. All policies retain valid input interpretation and explicit logging; experimental variants do not enter participant sessions on the basis of this table.

Policy role	Decision authority	What the comparison can test
Constrained Jev with local feedback	Local eligible action; provider chooses wording and speak/wait; qualified local feedback can bypass a new request.	Reference behavior and delivery delay under the shared scheduler.
Jev-led with local overrides	Provider considers meaningful eligible actions using state/history; local geometry, qualification, and final admission remain binding.	Whether broader choices improve sequence appropriateness without unacceptable delay.
Provider-led replay probe	Provider receives broad state and action choices; overrides are recorded as shadow checks rather than used to declare the model correct.	Unsupported choices and response timing offline. This is not an approved participant policy.

The ordinary inspected request contract accepts two to five candidate phrases. A full-eligible-set experiment therefore requires an explicitly versioned test adapter when more than five candidates are eligible; this draft does not claim that the inspected request wrapper supports an arbitrary catalog size. Each pilot will log its actual candidate count, serialized payload size, request path, and

model identifier. Small, medium, and full sets will be defined before execution rather than assigned favorable labels after observing results.

Paired execution will hold input timestamps, target geometry, validity, audio availability, and scoring rules constant. Model repetitions and execution order should be prespecified to expose stochastic variation and service-time effects. Independent request calls do not eliminate order-related network or server-load variation. Results will distinguish per-call transport measurements from whole-sequence behavior and will not substitute an average request latency for confirmation delivery time.

F.8 Metrics and Episode Definitions

For each policy, the report will summarize qualification-to-software-playback latency with median, 95th percentile, maximum, and count beyond a stated engineering threshold. The threshold must be declared before comparison and distinguished from a perceptual or clinical criterion. Episodes, rather than repeated requests within an episode, will be the denominator for delayed confirmations.

Jev request latency, failures, timeouts, calls per trial, cancellations, and late responses will be reported separately. A provider timeout can lead to a timely local fallback, while a successful provider response can still arrive too late for the current state. Provider reliability and delivered guidance reliability therefore require different measures.

Wrong-direction and stale hints will be labeled only when the available evidence shows that the instruction is unhelpful toward the task; a different but valid phrase is not automatically wrong, and missing evidence remains unknown. Judgments about repetition, unnecessary speech, excessive silence, and interruption appropriateness likewise require context beyond a sign check on an angle.

F.9 Illustrative Failure Analysis

An ordering failure can occur when a correction is queued before the controller reaches the target, and several distinct mechanisms could produce it. If playback checks use the older snapshot, the correction can play immediately before a pointing acknowledgment. Alternatively, the controller hold may not yet have qualified, leaving the ordinary-direction policy active during the hold, or action priority may admit a lower-priority cue despite newer pointing evidence. These causes require different fixes, yet they cannot be distinguished from the spoken text alone.

The intended correction is to suppress contradictory away-directions as soon as valid controller-on-target evidence is present, while retaining the independent qualification timer for the spoken pointing acknowledgment. A final playback check must repeat the suppression after any asynchronous wait or audio fade. If the controller leaves the target before qualification, no pointing acknowledgment should be emitted merely because the earlier request was valid.

A separate reference-frame problem arises when an object appears higher than a highlighted neighbor but the instruction says to look lower. If the instruction is calculated relative to the current eye ray, which could lie above both objects, the phrase can be geometrically consistent yet confusing because the highlight suggests a different reference object. Diagnosis therefore requires the recorded gaze ray, object positions, highlighted candidate, and playback time; it must not rely only on the review camera's screen-space layout.

These examples are diagnostic categories drawn from the development history. This appendix does not report their incidence or claim they have all been eliminated in a frozen participant build. The acceptance matrix in Appendix C defines evidence required to make those claims.

F.9.1 A Proposed Failure Taxonomy

Table F.3 classifies potential faults by where evidence is lost or acted on incorrectly. These categories organize review; they are not observed error rates. More than one category may contribute to an episode. When the trace does not establish a causal stage, the result remains unresolved rather than being assigned to Jev by default.

Table F.3: Proposed sequence-review taxonomy. A label requires evidence from the relevant stage, and does not imply a frequency estimate.

Category	Diagnostic evidence	Example distinction
Observation or geometry	Validity, source age, coordinate frame, gaze ray, object transforms	Wrong spatial input versus a correct cue calculated from that input.
State or qualification	Gate timers, transition cause, controller hit, vicinity winner	Immediate pointing evidence versus qualified pointing acknowledgment.
Decision or context	Exact request snapshot, candidates, history, response	Unsupported choice versus an admissible alternative wording.
Scheduling or delivery	Cancellation, queue, final check, clip readiness, playback start/end	A valid old proposal that becomes stale before it speaks.
Interaction or reference	Visible highlight, task state, phrasing, recent speech	A geometric direction that is confusing relative to the highlighted neighbor.
Recording or scoring	Clock mapping, missing events, join version, episode identity	Missing evidence or duplicate counting versus a demonstrated runtime fault.

F.9.2 Worked Sequence Diagnosis

The following cases are constructed examples, not participant observations. Their purpose is to show what evidence would justify a finding. Let t_q be the first supported qualification and t_p the actual software playback start, as defined in Chapter 5.

1. **A direction becomes stale during a controller hold.** A downward cue is eligible at request time. Before its response arrives, the tracked controller ray hits the target. The hold is not yet qualified, so a pointing acknowledgment would be premature. Under the intended policy, the search direction is suppressed and the hold gate preserved. If the old direction actually plays, the finding concerns admission or scheduling; if it was merely proposed and rejected, it is not a delivered wrong hint.

2. **Gaze qualifies while the controller remains elsewhere.** Valid winning-vicinity evidence satisfies the gaze gate at t_q. A looking acknowledgment is appropriate; a pointing acknowledgment is not. For a policy that routes this acknowledgment through the provider, repeated model waits followed by playback at t_p form one delayed gaze-confirmation episode. Under the reference local-feedback path, a comparable delay would instead require investigation of scheduling, audio readiness, or delivery. Attribution requires the wait and scheduler records. It cannot be inferred from the final spoken text alone.

3. **The eyes and a nearby highlight imply different reference points.** The gaze ray is above both a target and a highlighted distractor, but the target is higher than that distractor. A lower cue may be consistent with the gaze ray while appearing wrong relative to the highlight. This warrants a reference-frame or wording concern, not an automatic wrong-sign label. The review must inspect the state at playback rather than only a screenshot after movement.

For each reported real example, the completed thesis will provide the case-selection rule, a short surrounding timeline, the proposed versus played wording, the governing evidence, and the reason for the label. A few selected examples can explain a mechanism; they cannot establish its prevalence. Frequency claims require the defined evaluation denominator, including unknown and unreviewable episodes.

F.10 Interpreting Two-Second Delays

A two-second delay can contain several different intervals: waiting for qualification, waiting for a model response, an explicit model decision to wait, queued speech, audio readiness, and playback scheduling. Removing a provider wait does not establish that all delays disappear. Conversely, a one-second controller hold is an intentional qualification period, not necessarily a one-second processing fault.

The appropriate test starts from the first supported qualification for the particular event and ends at its actual software playback. A gaze event with a 100-ms gate and a controller event with a one-second hold must not share an undifferentiated origin. The report will retain both first evidence and qualification time so that readers can distinguish intentional gating from added delivery delay. Without a new verified sequence dataset and a wearer or acoustic measurement where required, this draft cannot claim that the roughly two-second delays have been removed.

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